Running Toward the Clinic: Speed, Biological Risk and Health in the AI Transition

A plain guide to what gets faster, what gets more dangerous, and why the two cannot be added up yet (illustrated edition)

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This is the plain-language companion to the technical paper *Running Toward the Clinic: Speed, Biological Risk and Health in the AI Transition*. Everything here comes from that paper, its appendix and its supplement. Where the paper is unsure, so is this guide. Figures marked *illustrative* use the paper's own illustrative values to show a shape.

Part 1 — The question: the same speed buys the medicine and the risk

Suppose AI keeps getting better at the kind of science that finds new medicines. That is worth wanting. A drug that reaches patients in 2031 rather than 2033 saves the people who would have died in those two years waiting for it, and they are real people, countable in principle, even if nobody has counted them.

Now suppose the same improvement also makes it easier for someone who wants to cause harm with biology. Not certainly. Not by an amount anyone has measured. But not obviously zero either, and the people who work on this for a living do not treat it as zero.

And suppose, third, that hospitals keep being attacked by criminals with software. Emergency departments divert ambulances. Scheduled treatment slips. Some of the benefit the first supposition bought never reaches the patient it was meant for, because the door it had to come through was jammed by the third.

The question this paper asks is what happens to the health of a population when all three run off the same clock. Faster capability brings the medicine closer. It may also bring the danger closer. And it does nothing at all about the jammed door.

Most writing about AI and risk takes one of those three and argues about it. The interesting difficulty — the one that took a paper to work through rather than an essay — is that they share a cause. You cannot buy the first without buying some amount of the second, and the third sits across the path of delivery whatever you do about the other two.

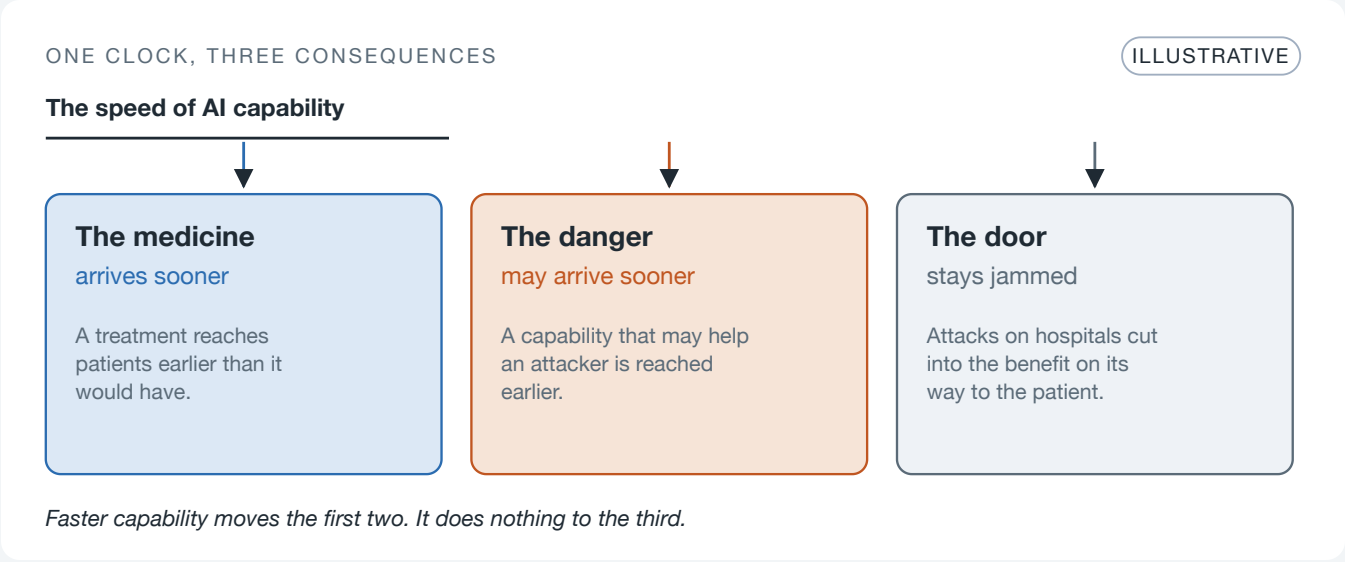


Figure 1. All three channels run off one clock, which is why they cannot be argued about separately. Speed moves the first two. It does nothing at all to the third. A drawing of the argument, not of data.

This is a sequel, and it says so

An earlier paper of mine, *Running Through the Rain*, asked a narrower version of the same question: should we welcome faster AI when the capability that helps defenders also arms attackers in cyberspace? It built four separate models, declined to add them up, and reached a cautious conditional answer. It counted one kind of harm against the benefit.

This paper counts three. Deliberate misuse of biology. Accidents and incidents at the laboratories doing the work. And attacks on the health systems that deliver the benefit. All three in the same unit as the benefit itself, which is the move that makes them comparable and also the move that creates most of the trouble.

The honest headline is that counting them does not settle the question. What counting does is show *why* it is not settled — which conclusions survive every assumption you might make, which ones flip depending on an assumption, and exactly which unmeasured quantity decides each flip. That is a different product from a number, and in this case it is the only product the evidence supports.

If you read the earlier paper, three things here are new. The benefit now has a delivery path that can be attacked, not just a date. The danger now has two incompatible descriptions that the evidence cannot choose between, and the paper carries both. And the central result is not about speed at all; it is about which of the paper's own conclusions depend on choices nobody can make yet.

Why "run or wait" is the wrong shape for this question

The earlier paper's image was someone caught in a downpour deciding whether to run to the far curb. Running means less time getting wet. The curb is where it is. The rain falls at the rate it falls. The only question is your speed.

That picture fails here twice over.

It fails once because the far curb is not fixed. The medicine does not arrive when the science is done. It arrives when the trials are done, the regulator has looked at it, the factory has made it, and the clinic has it on the shelf and a nurse who

knows to offer it. Speed in the laboratory moves the first of those and leaves the rest where they were. A capability that cuts the discovery stage from four years to one has not cut the wait from twelve years to nine unless everything downstream was waiting on discovery, and mostly it was not.

It fails a second time because running may thicken the rain. If the same capability that shortens the science also shortens the distance to a capability someone could misuse, then going faster does not only shorten your exposure. It can raise the rate you are exposed to. In the rain image, the runner's own speed changes the weather — which is not a thing rain does, and is the point at which the metaphor stops earning its keep.

So the paper does not ask "run or wait". It asks a narrower and more answerable question: under what conditions does moving faster leave a population with more healthy years of life, and under what conditions fewer? Those conditions turn out to depend on quantities nobody has measured, and — this is the part that surprised me while writing it — on three modelling choices that no amount of further searching will settle, because they are not gaps in the evidence. They are decisions about what to write down.

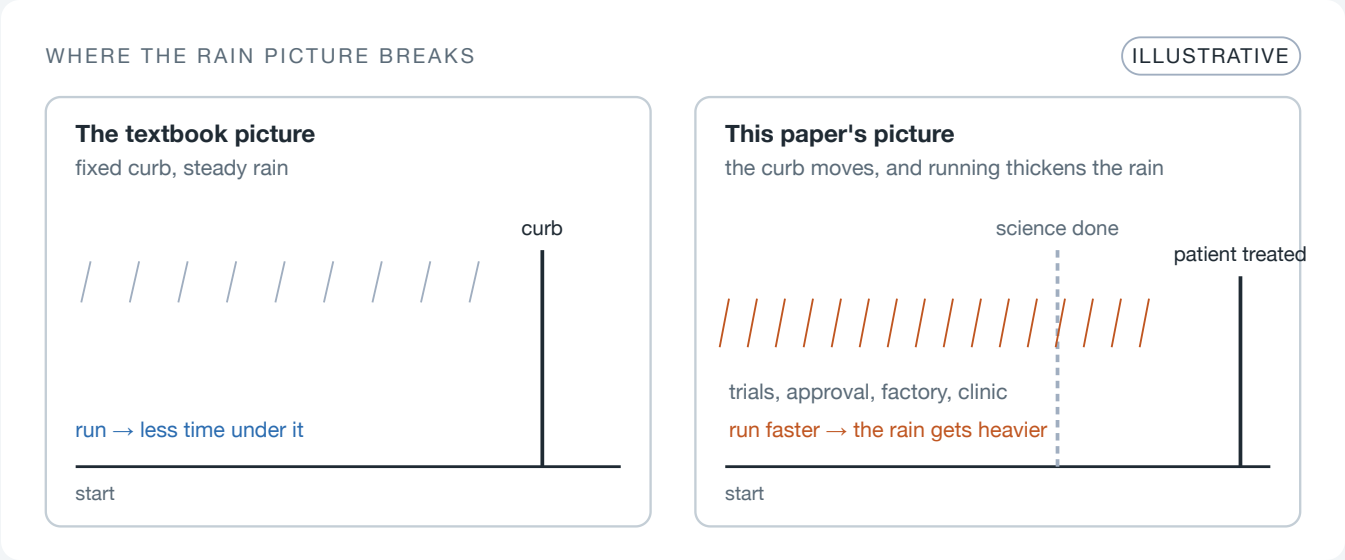


Figure 2. The analogy fails twice. The far curb is not where the science ends — it is where the clinic begins. And the runner's own speed changes the weather, which is not a thing rain does. A drawing of the analogy, not of data.

A benefit that arrives early and ends early

Here is the first result in the paper that most people, including me, get wrong on first acquaintance. It is worth doing slowly, because the rest of the paper's arithmetic has the same shape.

Suppose a new treatment, once it reaches patients, prevents fifty thousand years of lost health every year it is in use, and suppose it remains the best available option for ten years before something better replaces it. Call that a fixed-duration benefit: the length of the benefit is a property of the treatment, not of when it showed up.

Now accelerate. The treatment arrives in 2031 instead of 2033.

Count it up. Arriving in 2033, it runs from 2033 to 2043 and delivers five hundred thousand healthy years. Arriving in 2031, it runs from 2031 to 2041 and delivers five hundred thousand healthy years. The window slid two years earlier. It did not get longer. The total is identical.

That is the result: **earlier arrival of a fixed-duration benefit, counted in full, is worth nothing.** Not "worth little". Worth zero, exactly, in a ledger that adds up healthy years without weighting them by date. The intuition that earlier must be better is importing something the arithmetic does not contain.

Two things can make earlier arrival worth something, and the paper is careful about which is which. The first is discounting — if a healthy year in 2031 counts for more than one in 2041, then sliding the window forward raises the total, because it moves benefit from the lightly weighted end to the heavily weighted one. The second is a cutoff: if you only count what happens before some horizon, say 2040, then the 2033 arrival loses its last three years to the edge of the window while the 2031 arrival loses only one. Arriving earlier then fits more of the benefit inside the frame.

Both of those are real. Neither is "earlier is better" as a free-standing fact. And the distinction matters for policy, because the two repairs point at different levers:

discounting is an assumption you argue about, while the horizon is a choice about what to count that someone made when they set up the problem.

The paper proves a companion point that cuts the other way from the cheerful reading. Under discounting, the gain from making a candidate treatment *arrive sooner* dominates the gain from making it *more likely to reach patients at all* — not the reverse. If you expected the probability channel to be the more valuable one, the algebra disagrees, and Part 4 shows where.

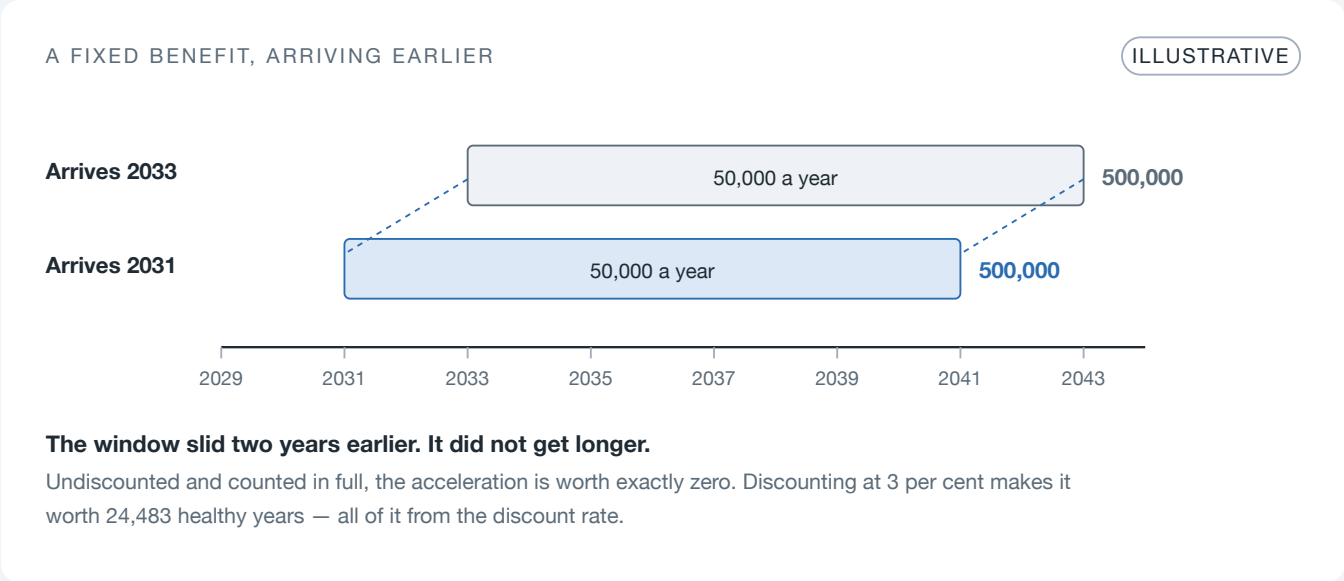


Figure 3. Both bars total the same. Earlier arrival moves the benefit's end forward along with its start, so in a ledger that weights every year equally the acceleration buys nothing. Illustrative values chosen to show the shape.

The occasion: one laboratory, as its owner describes it

The paper has a concrete occasion, and it is worth knowing about before the abstractions start.

A frontier AI developer has built itself a biology laboratory. That the laboratory exists is independently reported — investigative journalism from two outlets, confirmed by the company. Its size, its staff and its opening date are not stated anywhere.

Everything else about it is known only from the company's own account, and the paper labels it that way on every mention. I will do the same here, because the distinction between "reported" and "verified" is the whole of what can honestly be said.

By the company's account: the laboratory works on fundamental biology rather than drug discovery specifically, and runs no clinical work for now. It operates only at the two lowest biosafety levels. It handles no pathogen that can infect humans. All bench work is done by human scientists, and model-directed control of laboratory equipment is an aim rather than current practice. The company's wider life-sciences unit does include drug discovery by its own description, and nothing the paper read establishes where that work happens.

If those statements hold, the laboratory sits outside the population that the high-containment biosafety literature studies, which is maximum- and high-containment facilities. That placement rests on the company's account alone.

And here is the part that should be read slowly. **No independent inspection, permit, registration or institutional biosafety committee record was located.** Whether the operating entity falls inside the US research-oversight regimes is not determined on the record the paper has. The company's statements bear on the first of two triggers for US oversight — possession of a listed agent — and nothing read establishes the second, which is federal life-sciences funding.

So the honest summary is: a laboratory exists, it is owned by a company whose models are the subject of the risk debate, and what it does is attested by that company and by nobody else.

Four conditions, reported one at a time

The literature on AI-directed laboratories names one mechanism by which such a facility could lower the barrier of hands-on skill that misuse would otherwise require, and says plainly that the test has not been done.

Carrying that concern to any particular facility needs four conditions. The paper reports each singly, as the company reports it, and explicitly does not combine them into a conclusion.

1. **Biological rather than chemical work.** Met.
2. **Automation that substitutes for skilled hands rather than supervising them.** Unmet, as the company reports it — bench work is human-performed and model direction is an aim.
3. **Acceptance of experiments specified by outsiders.** Unknown. Nothing the paper read addresses it at all.
4. **Work with agents anywhere near those this literature concerns.** Unmet, as the company reports it.

Two of those four rest on the company's own statements with no independent verification located, and one is simply blank. Reporting them one at a time rather than multiplying them into an overall verdict is deliberate: four conditions reported singly is what the evidence supports, and a product of four numbers, two of them self-reported and one unknown, is not.

If bench work is human-performed as reported, then the specific route this literature names does not apply to that work. That would not establish the absence of AI assistance, or of every effect on hands-on skill. It would establish that one named mechanism is not the one operating.

The paper is also explicit about a question it does *not* ask: whether this laboratory is safe, or safer than a public one. Nothing it located can answer that, and no company-against-public facility safety comparison exists in anything it read. What it asks instead is a question about distances — whether a developer-owned laboratory, taken exactly as reported, moves the capability a countermeasure needs and the capability misuse needs, and if so, whether the net effect on misuse loss is up or down. That question turns out to have no determinate answer either, for the reason Part 3 gives: the answer depends on what "brings a capability closer" is taken to mean.

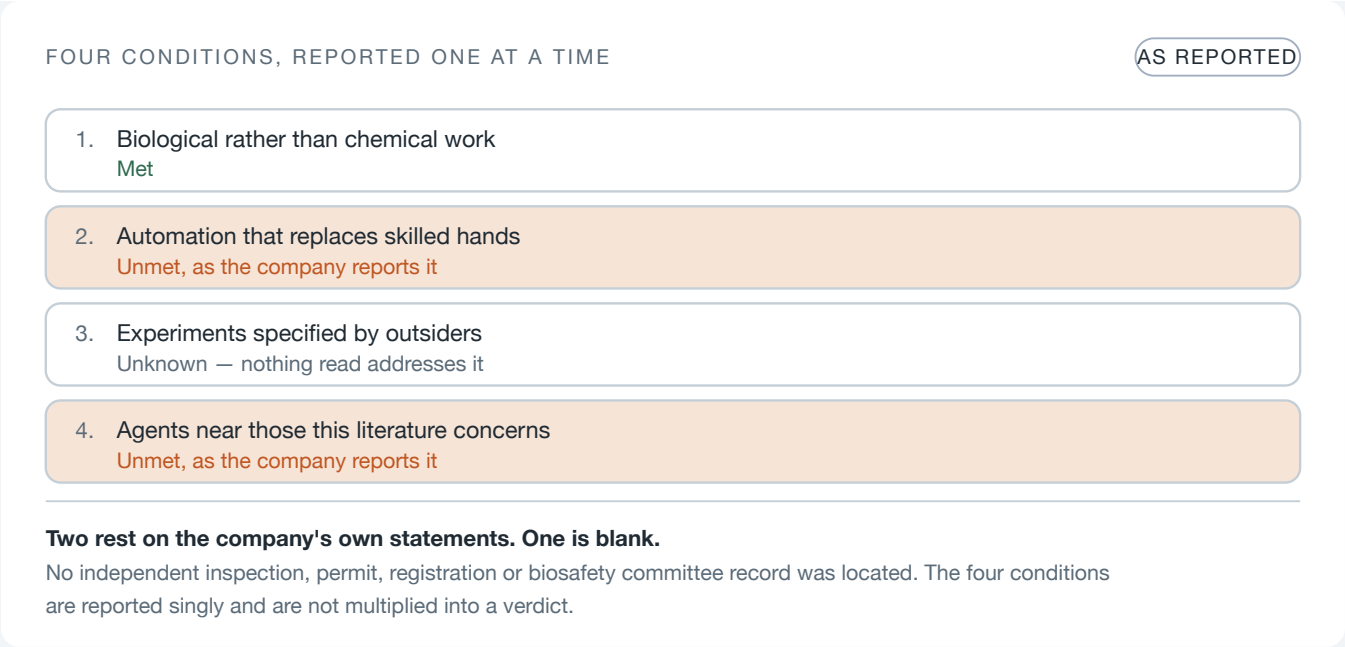


Figure 4. Each condition is reported singly, as the company reports it. Multiplying four values together — two self-reported, one unknown — would produce a number the record does not support. Reported facts and stated absences; no verdict.

What this guide is, and is not

This guide retells the working paper of the same name for readers who do not work in political science, epidemiology or AI. It follows the paper's own order. It keeps the paper's hedges: where the paper is unsure, this guide is unsure in the same places and for the same reasons, and where the paper found it had overclaimed and corrected itself, the correction is here too.

It contains no detail about how biological harm could be caused, and neither does the paper. Every sentence about misuse in both documents stays at the level of "a capability that may help an attacker" and goes no lower. That is deliberate, and it is a real limit on both: a reader might reasonably want to know how large the danger is, in what form, from whom, and neither document will tell them. What the paper can say about the evidence — what has been tested, by whom, with what power to detect anything — is in Part 7.

Two warnings about what you will not find.

There is no verdict. The paper produces no number for whether faster is better or worse for health. It proves which conclusions hold whatever you assume, which ones flip, and which unmeasured quantity decides each flip. If you came for a recommendation, the paper declines to give one, and Part 9 argues that declining is the honest answer here rather than a dodge — including against the obvious objection that one should act on the worse case when the balance is unknown.

There is no ranking of the things one might do about it. The paper discusses several levers: faster capability, countermeasures that arrive sooner, hardening the hospitals, slowing down. It does not say which is best. Saying so would require exactly the numbers it has just finished demonstrating nobody has.

How the ten parts fit together

The shape is simple, and knowing it in advance makes the middle easier.

Parts 2 and 3 build the apparatus: one unit for everything, and the three choices the paper refuses to make for you. Part 4 is the good news — what holds whichever way those choices go. Part 5 is the bad news — what depends on them, and on which one. Part 6 takes the single most important conditional result apart in slow motion, because it is the one a policy argument would lean on.

Parts 7 and 8 leave the mathematics and look at evidence. Part 7 asks what has actually been measured about the misuse channel and finds the chain of inference breaks at the third of five links. Part 8 looks at the one channel that does have a measured causal magnitude — attacks on hospitals — and shows why having a number there, and only there, is a trap rather than a comfort.

Part 9 is the refusal, stated and defended. Part 10 is the list of what would settle what, which is the closest thing to a conclusion the paper has, and the part a reader with a research budget should start from.

Part 2 — One yardstick: healthy years of life, each counted once

You cannot compare a medicine arriving two years early with an attack on a hospital unless the two are measured in the same thing. Most arguments about AI and risk never get this far. They put the benefit in one currency — lives, cures, productivity — and the harm in another — probability of catastrophe, incidents, breaches — and then compare them by tone of voice. Whoever sounds more serious wins, and nothing has been established.

This paper insists on one unit for everything it counts. That single decision is responsible for both its useful results and most of its difficulty.

The unit is a year of healthy life

The unit is the one public-health bodies already use: the **disability-adjusted life year**, or DALY. One DALY is one year of healthy life lost.

It has two parts, and the split matters later. A death costs the years the person would otherwise have lived — a death at forty costs more than a death at eighty, because there was more life left to lose. An illness that leaves someone impaired without killing them costs a fraction of each year it lasts, with the fraction standing for how much of a healthy year that condition takes away. Add the two and you have the burden of a disease, or of an event, or of a decision, in one number.

Nobody in the field claims the unit is perfect. It requires saying how much a year lived with a given impairment is worth relative to a year lived without one, and reasonable people disagree about those weights, sometimes sharply. It treats a year as a year regardless of whose it is, which some find obviously right and others find obviously wrong.

But it has the two properties this paper cannot do without. It is additive, so losses from different causes and different people can be summed. And it is already the

unit in which both the health gains from new medicine and the losses from disasters get estimated, so using it requires no new conversion factor invented for the occasion. The World Health Organization's own valuation is used throughout, unchanged, precisely so that no result depends on a choice made here.



Figure 5. The unit is additive and already standard in both health economics and disaster estimation, which is why the paper can use it without inventing a conversion. A definition, not a result.

Why one unit is doing real work here

Put the four things the paper counts side by side and the point becomes concrete.

A medicine that arrives earlier saves healthy years. Someone made ill, or killed, by deliberate misuse loses healthy years. An accident at a laboratory doing the work costs healthy years. An attack that shuts down a hospital's systems costs healthy years among the patients who were already in its care and the ones whose new treatment never arrived. Same unit, four sources, one ledger.

Without that, the argument degenerates in a specific and familiar way. One side says the medicine is worth it. The other says the risk is unacceptable. Because the two quantities are not denominated in anything shared, both statements can be true at once and neither can be checked against the other. Put them in one unit and the disagreement becomes a disagreement about numbers — and numbers

can be found, or shown to be missing. Most of this paper consists of showing which are missing, and that is only possible because everything was written in the same currency first.

There is a cost to the move, and it should be said plainly. Writing a potential pandemic and a ransomware outage in the same unit invites the reader to think the two are equally well understood. They are not, and Part 8 is about exactly that asymmetry.

Six places a year of health can change, and one dot each

Once everything is in one unit, a new way to be wrong appears: counting the same lost year twice.

The paper handles this by laying out every route by which the capability can change someone's health, and requiring each route to be booked in exactly one place. There are six.

1. **The benefit.** Healthy years gained because a treatment reached patients.
2. **Harm to care already being delivered.** A patient in the hospital today whose treatment is disrupted.
3. **Harm to the new therapy's delivery** — in two forms: the doses that never got given during an outage, and the trials that ran late because the systems running them were down.
4. **Deliberate misuse.**
5. **Events at the facilities doing the work** — accidents and incidents, as distinct from misuse.
6. **Countermeasures and remediation** — what it costs to prevent and to clean up.

The rule is the whole of the bookkeeping: **every year of healthy life gained or lost is booked in exactly one of those six.** The paper draws this as a grid, with a dot in the one cell where each route is counted — and, next to several of the dots, a cross

marking the tempting second place the rule excludes. The crosses are the useful part. They are where a careful person would otherwise double count.

Two of them are worth naming. Deliberate outbreaks and accidental ones are counted on the same consequence scale, because an outbreak's health cost does not depend on whether someone meant it; but each event is assigned to misuse or to facilities, never to both. And a harm counted as an attack is not also subtracted from the benefit, even though it is perfectly true that the attack reduced the benefit — the reduction is already in the attack's own column.



Figure 6. The crosses are the useful part — they mark where a careful person would otherwise count the same lost year twice. The rule is an accounting identity, true by construction rather than observed.

A worked double count, and the rule that catches it

The clearest case is an attack on a hospital, and it is worth walking through, because this is where the rule earns its place.

An attack takes a hospital's systems down for eleven days. During that window two kinds of patient suffer. There are the patients already under the hospital's care, whose treatment is delayed or degraded. And there are the patients who

should have begun the new treatment — the one the faster capability bought — and did not, because the service that would have started them was offline.

So the health loss from that attack has two components, and the paper writes it as exactly two: the baseline care that was lost, plus the new benefit that was foregone. Nothing else.

Now the trap. It is natural to reason as follows: the attack cost eleven days of baseline care, and *also* the new treatment was delayed by eleven days, so let us subtract eleven days of benefit from the benefit column too. Do that and a patient who died during the outage has now been counted twice — once in the hospital's lost care, once as benefit that never arrived. The total harm comes out too large, and the direction of the error is not random: it inflates the harm side, which is the side the whole argument is about.

The rule forbids it by allocating every unit of patient-time to exactly one attack and every patient to exactly one of the two components. A patient who dies during an outage is counted in the baseline-care loss or in the foregone benefit. Never both.

This sounds like housekeeping, and it is. It is also the kind of housekeeping that quietly decides results. A ledger that double counts will show a harm larger than it is; one that drops a term will show it smaller; and either way the paper's central question — which way the balance tips — is being answered by the accounting rather than by the world.

Two honest qualifications. The rule is an accounting identity, not a finding: it is true because of how the ledger was built, not because of anything observed, and it would be true in a world where all the numbers were different. And the simple addition of the terms is an approximation that holds when the losses do not overlap and the changes being considered are small — two attacks on the same ward in the same week are not two independent events, and the paper's arithmetic does not pretend otherwise.

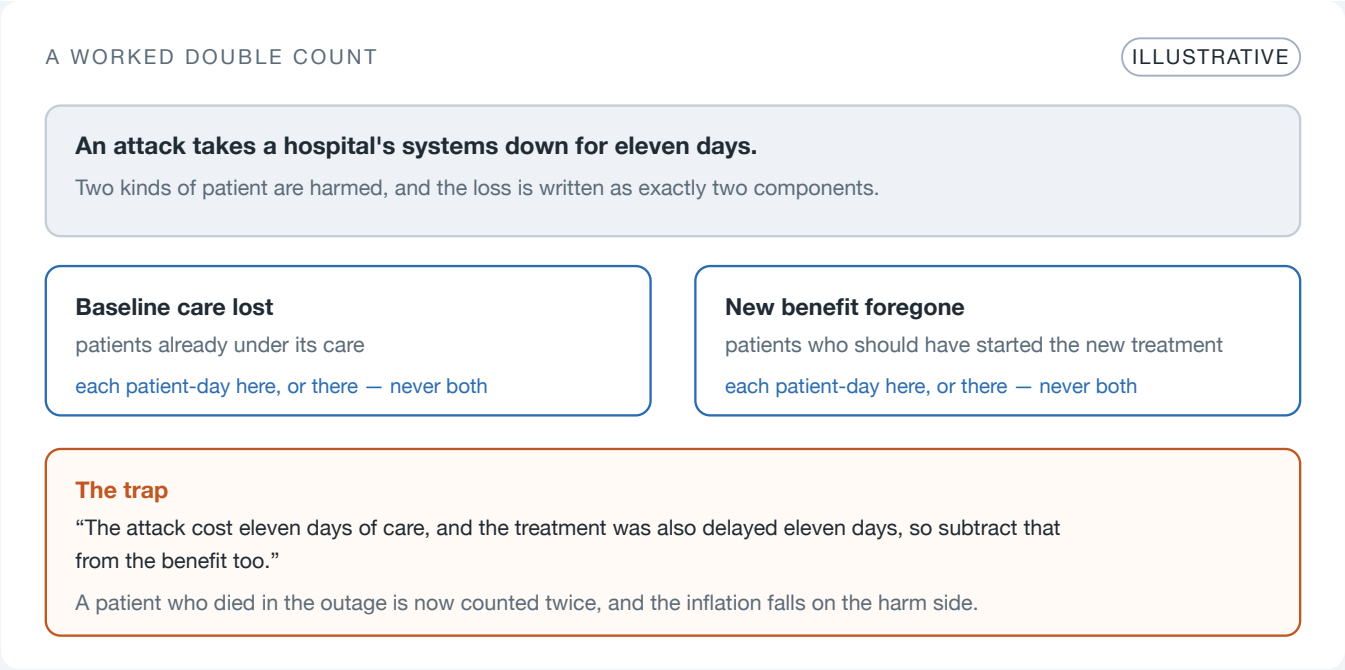


Figure 7. The error is natural, and it is not symmetric: it inflates the harm side, which is the side the whole argument turns on. Illustrative numbers chosen to show the mechanism.

The two sides are booked on different horizons

Here is a convention in the paper that I did not notice I had adopted until the result it produces was pointed out, and it leans one way.

The ledger counts over a window — from now to some horizon. Call it the year 2060 for concreteness. The benefit is counted only up to that horizon: healthy years the treatment delivers in 2065 do not appear, because they fall outside the frame.

Harms are counted differently. A death is booked, undiscounted, at the date it occurs, and it costs all the life-years it took away — including the ones that would have fallen after the horizon. A person killed in 2059 at the age of thirty loses the whole of their remaining life expectancy, not the single year that fits inside the window.

Both conventions are defensible on their own. Together they are asymmetric: a year of life *gained* after the horizon is worth nothing, and a year of life *lost* after the horizon is worth one. That asymmetry favours the harm channels, and it

favours them in the misuse term specifically, whose per-event cost sits on the same untruncated footing.

The paper states the convention rather than fixing it, and gives its reason: truncating the harm side would make each event's cost depend on the date it happened, which breaks the simple structure the rest of the results are built on. That is a real trade-off and not an excuse. But a reader comparing the two sides should know that the frame itself is not neutral, and the paper had not said so until a late pass through its own appendix turned the asymmetry up.

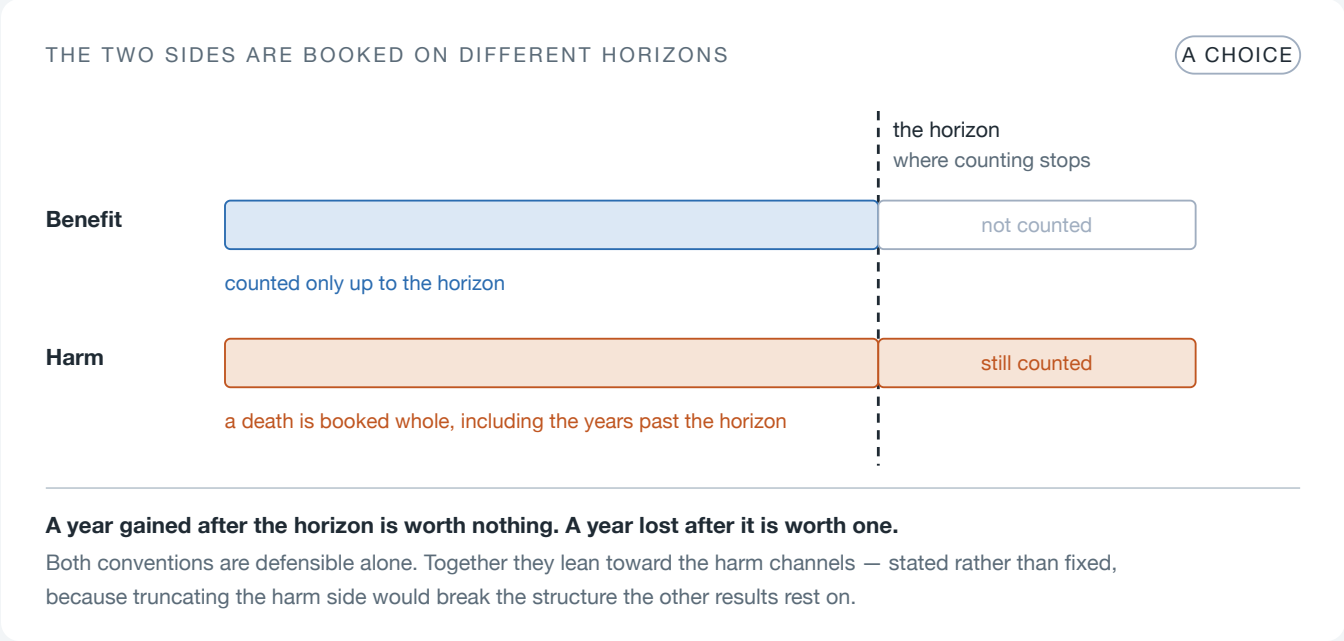


Figure 8. The frame itself is not neutral. The paper's reason for keeping it is that truncating the harm side would make each event's cost depend on its date, which breaks the structure the other results rest on. Schematic.

What the unit does not do

Being able to compare is not the same as knowing what to compare.

The paper can write every channel in healthy years. It cannot fill most of them in. How much health benefit faster AI actually produces is not measured in any source it searched. How much loss a single misuse event would cause is not measured anywhere it could find — not estimated badly, not estimated with wide bounds: not estimated. And how many healthy years an excess hospital death

costs depends on whether that death ended a life with decades left or advanced one that had weeks, a question the available data does not answer and whose two answers differ by a factor of about 260. That is Part 8.

So the unit is a frame, not an answer. It makes the question well posed. It does not make it answered. Anyone who tells you the sum comes out positive, or negative, is supplying numbers the evidence does not contain — and the remaining parts of this guide are, among other things, a guide to noticing when that is being done.

Part 3 — Three choices nobody can settle

Everything in Part 2 was bookkeeping. This part is where the paper's real difficulty lives, and it is not a difficulty about evidence.

Three times in building the model, I reached a fork where two descriptions of the world were both reasonable, both consistent with everything anyone has published, and led to different answers. I could not find a source that chose between them. More to the point, I do not think a longer search would have found one, because what is missing is not a measurement. What is missing is a decision about what to write down.

The paper therefore carries both sides of each fork, side by side, neither one as the default. That is unusual, it roughly doubles the work, and it is the single most important thing about the paper, because it is what lets the results be sorted into the ones that survive the forks and the ones that do not.

Choice one: is the danger set by how far you have got, or how fast you are going?

Picture a mountain road at night. There are two quite different ways to say what makes it dangerous.

One: the danger is the altitude. The higher you have climbed, the narrower the road and the longer the drop, and that is true whether you are crawling or racing.

Your speed changes how long you spend at each altitude but not how dangerous that altitude is.

Two: the danger is the driving. What threatens you is how fast you are taking the bends. Where you have got to on the mountain hardly matters.

The paper's first fork is exactly this, applied to misuse risk. Under the first description — the paper calls it **level-set** — the hazard depends on the capability level that has been reached. A more capable system is a more dangerous one, and progress matters because it raises the level. Under the second — **speed-set** — the hazard depends on the rate of advance itself: a field moving fast is a field where things go wrong, where oversight lags and where whatever safeguards exist were designed for last year's systems.

Both are defensible. The earlier paper, *Running Through the Rain*, used the speed-set form for its main results. This paper carries both, and the choice turns out to matter more than I expected.

Three results exist only under level-set. The most important is **truncation**: if the dangerous capability lies beyond what the horizon would ever reach at current speed, then going faster pulls it inside the window where it can be counted — which creates a cost out of nothing but acceleration. That mechanism needs danger to be indexed to a level, because it is about *arriving* at a level sooner. Under speed-set there is no level to arrive at, and the mechanism does not exist. Not "is smaller". Does not exist.

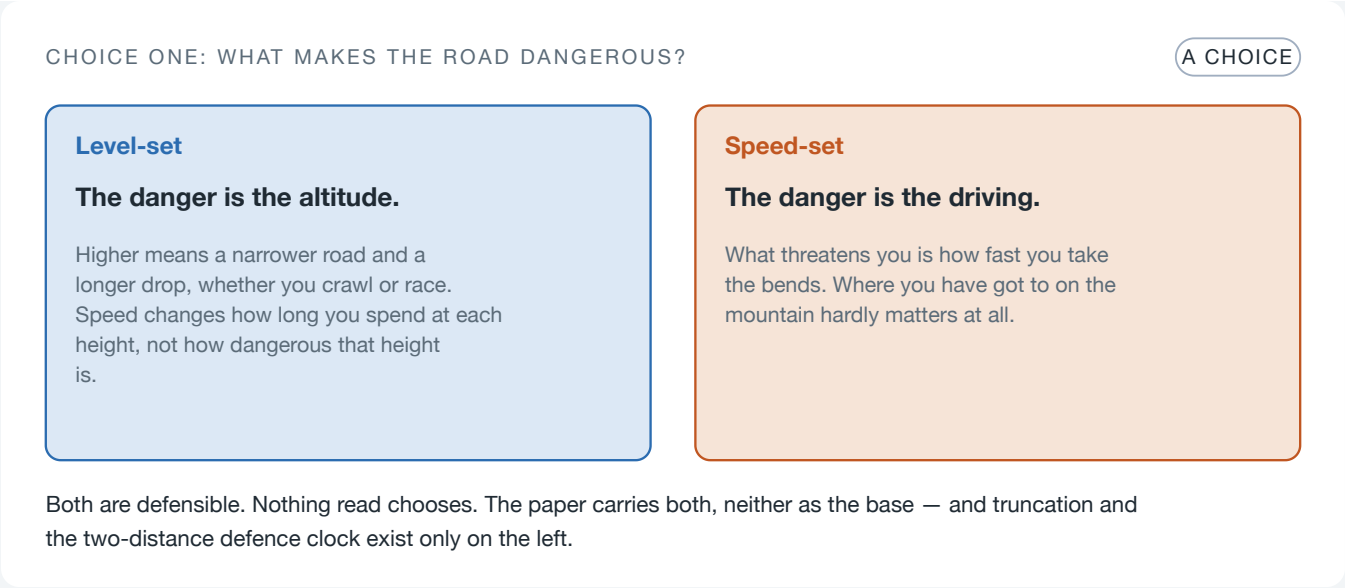


Figure 9. A modelling choice, not a measurement gap: no further search produces it. Three results in the paper exist under the left-hand reading only. A drawing of the choice, not of data.

Choice two: when does the clock start?

The second fork is separate from the first, and it is about counting rather than about mechanism.

A research programme sets out toward some capability. Under the paper's notation it reaches that capability at a date you can write down: the distance to be covered divided by the speed of advance. Before that date, the capability does not exist. After it, it does.

So when does misuse risk start accruing?

Under the **ramp-on** reading, it accrues all along. The programme is building toward something dangerous, and partial progress is partially dangerous — intermediate systems, intermediate knowledge, intermediate tools, all of it available before the finish line. Under the **arrival-on** reading, the clock starts at arrival. Before the capability exists, there is nothing to misuse, and counting risk from the programme's first day counts a danger that is not there yet.

Both readings describe something real, and which one you pick changes the arithmetic in a way that is easy to miss: it changes the *length of the exposure*, and length multiplies everything. Acceleration shortens the run-up to arrival — so under arrival-on, faster progress means the dangerous period starts sooner but the pre-arrival period, which was never counted, shrinks to nothing. Under ramp-on, acceleration compresses a period that was being counted all along.

The dependence is not decorative. Four of the paper's cases where the two hazard specifications give *opposite* answers about speed exist only on the ramp-on reading. Compare them on the arrival-on reading throughout and exactly one of the four survives. A reader who met the paper's "four disagreements" without knowing which reading produced them would be badly misled, which is why Part 5 states the reading before the count.

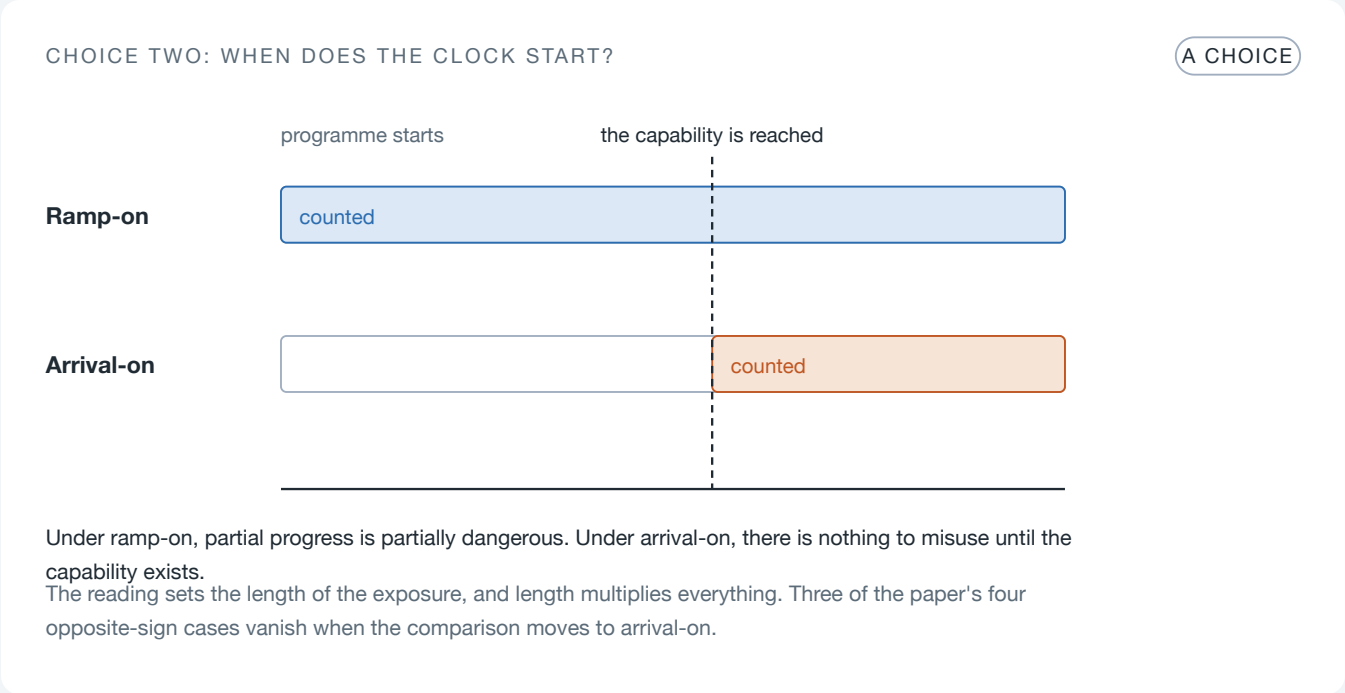


Figure 10. A separate axis from Figure 9, and again a choice rather than a gap. Schematic; the proportions are illustrative.

Choice three: what does it mean to bring a capability "closer"?

The third fork is the subtlest, and it is the one I would have got wrong without being forced to write it out.

Suppose a laboratory does work that brings a potentially misusable capability nearer — shortens the distance that would have to be covered to reach it. That sentence feels unambiguous. It is not. It can mean two different interventions, and under the level-set description they are not the same thing at all.

Convention A compresses the whole danger profile along with the threshold. The shape of the climb is preserved; the mountain is simply shorter. You reach the same summit, with the same hazard at the summit, at a lower capability level — and every waypoint on the way up is now more dangerous than the matching waypoint before, because the whole profile has been squeezed toward the start.

Convention B leaves the profile exactly where it was and moves only the threshold. The hazard at each capability level is what it always was. What changes is where you stop: the summit is now reached at a lower level than before. And if the hazard rises with capability — which is the usual assumption — then stopping lower means stopping at a *less* dangerous place.

Both conventions are faithful readings of "the facility brought the capability closer". Nothing in anything I read chooses between them.

The same sentence, opposite effects

Here is that fork with numbers in it, because the abstract statement is not believable until you see it happen.

Take the paper's own illustrative profile: the hazard rises as the square of how far along the path you are, scaled so that it equals 1 at the threshold. Start with the threshold at distance 1. Then the hazard at the quarter-way point, distance 0.25, is 0.0625, and the hazard at the summit is 1.

Now a laboratory halves the distance to the threshold, from 1 to 0.5. Same English sentence: *the facility brought the misuse capability closer.*

Under convention A, the profile compresses. The hazard is now the square of (distance ÷ 0.5) rather than (distance ÷ 1). At distance 0.25 — which is now the

halfway point rather than the quarter-way point — the hazard is 0.25 instead of 0.0625. Four times higher. The summit hazard is still 1. Every point on the road got more dangerous and the top of the mountain is just as high as it was.

Under convention B, the profile does not move. At distance 0.25 the hazard is 0.0625, exactly as before. What changed is that the climb now stops at 0.5, where the hazard is 0.25, instead of continuing to 1, where it was 1. The road is unchanged and you get off it four times lower down.

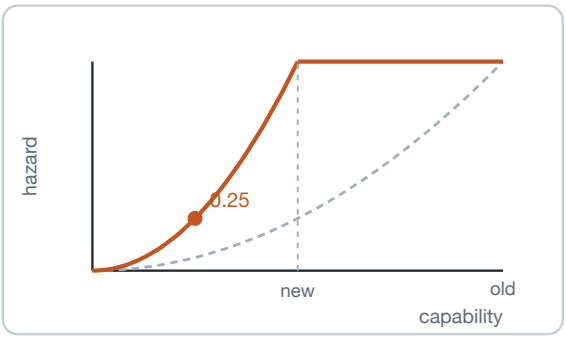
So: under A the intervention raised the danger at every point and kept the peak; under B it left every point alone and cut the peak to a quarter. Under the level-set description, on the ramp-on reading, and under conditions the paper states precisely, the same verbal intervention **raises misuse loss under A and lowers it under B.** Opposite signs, same sentence, same world, different decision about what the sentence meant.

This is the clearest instance of the paper's general predicament. It is not that we do not know how big the effect is. It is that we have not said what the effect *is* until we have made a choice that the evidence does not make for us.

CHOICE THREE: WHAT DOES "CLOSER" MEAN?

A CHOICE

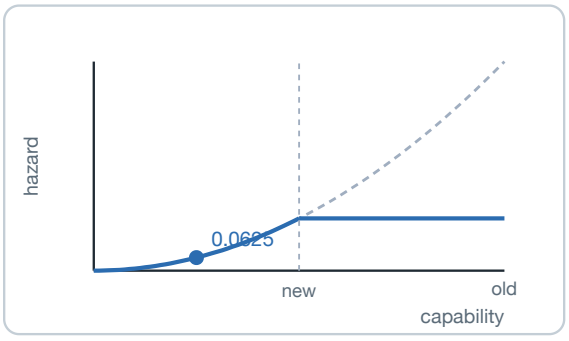
The same sentence: "the facility brought the misuse capability closer." The threshold halves, from 1 to 0.5.



Convention A — the profile compresses

Every point more dangerous; the summit still 1.

Quarter-way hazard 0.0625 → 0.25, four times higher.



Convention B — the profile stays put

Every point unchanged; you stop climbing lower.

Summit hazard 1 → 0.25, a quarter of its height.

Opposite signs, same sentence, same world — a different decision about what the sentence meant.

Figure 11. The clearest case of the paper's predicament: not that the size of an effect is unknown, but that the effect is not defined until a choice the evidence does not make has been made. The paper's own illustrative profile.

Why these are choices and not gaps

It matters to be clear about the difference, because a gap invites a research programme and a choice does not.

A gap is a number nobody has measured yet. The loss a single misuse event would cause is a gap: there is a true value, nobody has estimated it, and a well-designed study would move us closer. Part 10 is a list of gaps, and it exists because gaps are actionable.

A choice is different. Whether misuse hazard is set by level or by speed is not a quantity waiting to be measured; it is a question about which simplification to build on, and both simplifications are wrong in the way all models are wrong. Whether to count risk from the programme's start or from arrival is a question about what the counted window is for. Whether "closer" means a compressed profile or a moved threshold is a question about what intervention you are describing.

You could, in principle, settle some of this by building a richer model with both mechanisms in it and letting data choose the weights. The paper says plainly that this would need data that does not exist, and that proposing it as a solution to a problem of this kind is a way of deferring rather than answering. Meanwhile the honest thing is to carry both and label everything accordingly.

So when the rest of this guide says a result is conditional, it means something precise: not "we are unsure of the magnitude" but "the sign depends on a choice, and here is the choice".

What carrying both sides costs, and what it buys

The cost is real. Every level-set statement in the paper appears twice, once under each convention. Every claim about the misuse marginal is tagged with its exposure reading. The appendix carries parallel derivations where one would have done. A reader has to keep track of which branch they are on, and the paper's tables have a column for it.

What it buys is the only thing the paper actually delivers: a clean separation between the conclusions that survive all three forks and the conclusions that are artefacts of one branch.

That separation is the next two parts. Part 4 is the survivors — five results that hold whichever way all three choices go, each within its own conditions, plus the accounting identity from Part 2. Part 5 is the rest, each one labelled with the choice it depends on.

And there is a discipline hidden in the arrangement that I would recommend to anyone building something similar. Once you are carrying both branches, you cannot quietly pick the branch that flatters your conclusion, because the other branch is printed on the same page. Several results in this paper came out weaker than the first draft claimed, and the branch structure is why.

Part 4 — What holds whichever way you choose

Three choices, two ways each, and no way to settle any of them. You might reasonably expect nothing survives that. Something does, and this part is the good news in the paper.

Six things hold the same way whichever way all three choices go. Five are results about mechanism. The sixth is the bookkeeping identity from Part 2. Each comes with its own conditions, stated here and stated more precisely in the paper, and a result that holds "whatever you choose" still does not hold unconditionally — the conditions are about the shape of the hazard and the position of dates inside the counted window, not about the three forks.

Once the capability is there, going faster is worse

Take the level reading for a moment, where danger depends on how far the capability has got.

A capability advances to some dangerous level and then, for this particular danger, stops mattering — not because progress stops, but because beyond that point further progress adds nothing to *this* hazard. Before the threshold there is a ramp. After it, a plateau.

Speed shortens the ramp. It does not shorten the plateau. It lengthens it, by exactly the time it saved on the ramp, because the window you are counting over has not moved.

Work it through with the paper's illustrative profile — danger rising as the square of how far along the path you are, equal to 1 at the threshold, counted over a thirty-year window compressed here to six model years.

At the slower speed, the threshold is reached in year 2. The ramp contributes 0.67 units of accumulated danger and the plateau, running from year 2 to year 6, contributes 4.00. Total: 4.67.

Double the speed. Now the threshold is reached in year 1. The ramp contributes only 0.33 — half the speed, half the time climbing, and the compression is real. But the plateau now runs from year 1 to year 6 and contributes 5.00. Total: 5.33.

Accumulated danger rose by 0.67, about one seventh, *because* of the acceleration. The saving on the ramp was 0.33 and the extra plateau cost 1.00. Speed bought a third of a unit and spent a whole one.

That holds under both danger readings, each with its own condition on the hazard's shape, and under both answers to when the clock starts. It is also the one piece of the earlier paper's machinery that carries over to biology unchanged.

The lesson generalizes past the model. Acceleration helps only where there is something still to be compressed. Past the point where the dangerous thing

already exists, speed is spending without buying.

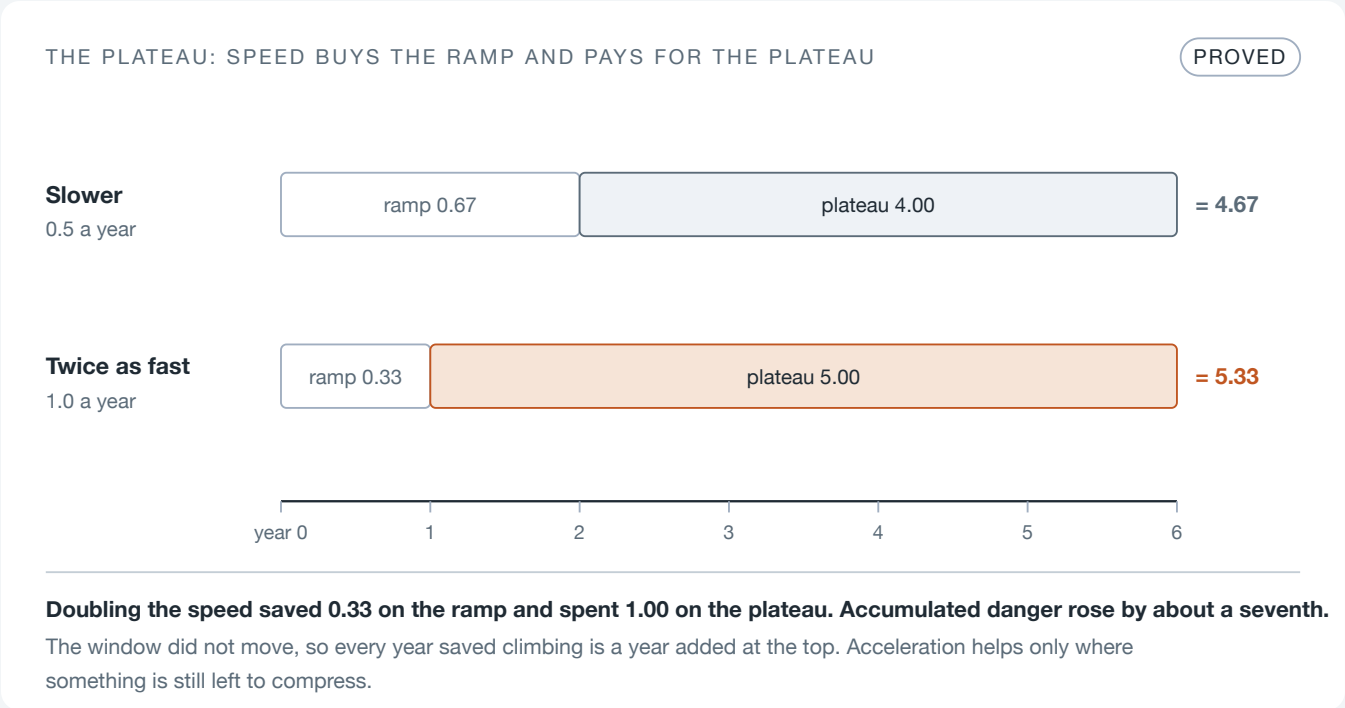


Figure 12. The saving on the ramp is real and smaller than the extra plateau it buys. This holds under both hazard readings, each within its own condition on the hazard's shape. The paper's illustrative profile; arithmetic checked.

Where the countermeasure waits on people, speed buys nothing

A countermeasure — a vaccine, a detection system, a screening requirement at synthesis providers — arrives when two conditions are met. Someone has to be able to build it, which waits on capability. Then it has to be approved, manufactured, distributed and actually taken up, which waits on institutions and people.

Speed compresses the first. It does nothing to the second.

The paper writes the marginal value of shortening the capability distance the countermeasure waits on, and it has a clean form: the share of the wait that capability sets, divided by the speed, multiplied by how much the countermeasure actually lowers the hazard when it lands.

Put numbers in it. Suppose the hazard on the plateau is 1 and falls to 0.2 once the countermeasure is deployed, and the capability is advancing at half a unit per year. On the branch where the countermeasure is waiting on capability, cutting its required distance by one unit lowers accumulated danger by 1.60 units. That is a large effect and it is the case for investing in the science.

On the branch where the countermeasure is waiting on adoption, the same calculation gives exactly 0.00. Not a small number. Zero, identically, because the term that carries the capability share has gone to nothing.

So there is a floor, and it is sharp rather than gradual. Shortening the capability distance helps while that wait binds, and the moment the binding constraint becomes approval or uptake, further cuts buy nothing at all. Crossing that line is a change of regime, not a weakening of the result.

This is the most directly useful thing in the paper for anyone deciding where to spend. If the countermeasure for your scenario is already technically feasible and stuck in regulatory review or in distribution, then making the underlying science faster is not a weak lever. It is not a lever.

The paper is careful about one extension here. The clean form assumes the countermeasure lands as a step — before it, one hazard; after it, another. Real coverage ramps up. The same form survives if the ramp moves rigidly with the deployment date and finishes inside the counted window, and it can fail if coverage is still rolling out when the window closes. That qualification was missing from an earlier draft and was added after the formal checks found it.

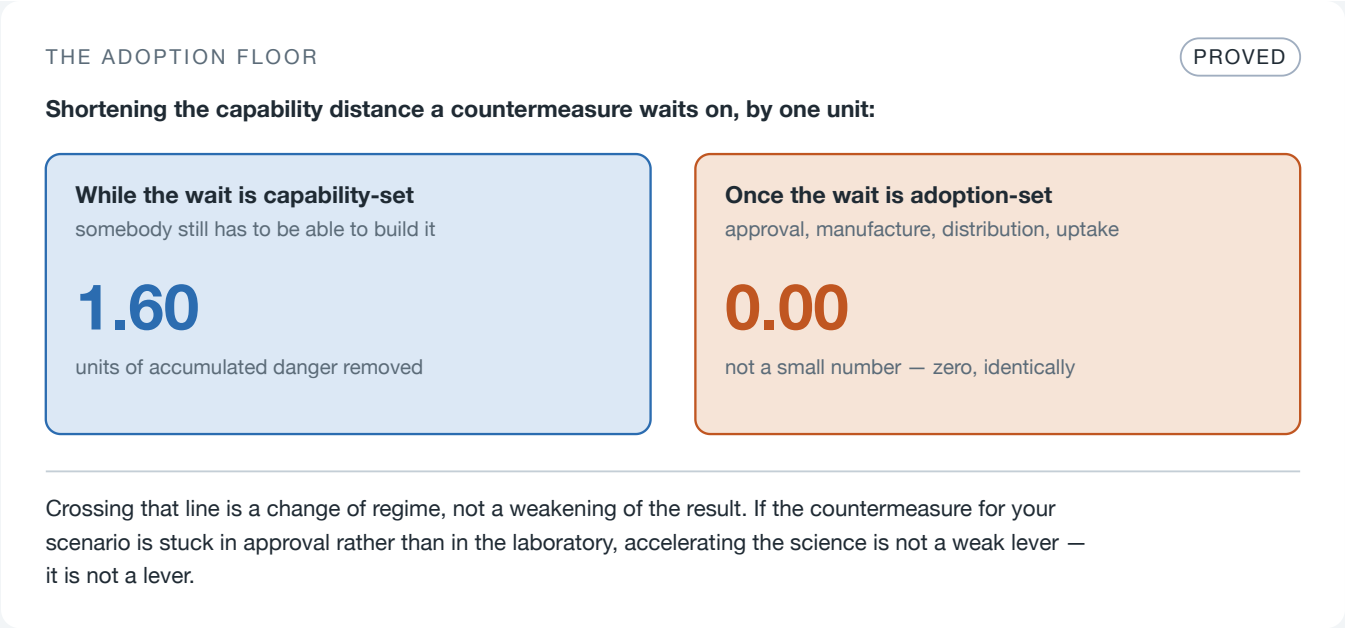


Figure 13. The most directly useful result in the paper, and the sharpest cut-off. Illustrative parameter values; the zero is exact, not rounded.

Arriving earlier is worth nothing on its own

Part 1 showed this one with a treatment arriving in 2031 instead of 2033: fifty thousand healthy years a year for ten years, five hundred thousand either way, the window sliding without lengthening. Counted in full with nothing cutting it off, earlier arrival of a fixed-duration benefit is worth exactly zero.

Now add the repair, with numbers, because the size of it is the interesting part.

Discount at three per cent a year — a standard rate in health economics, chosen here only because it is standard. The same benefit, arriving at once, is worth 426,510 healthy years in present-value terms. Arriving two years later, it is worth 402,027. The two-year acceleration is worth 24,483 healthy years.

That is a real quantity and nobody should dismiss it. But notice where every unit of it came from. The undiscounted totals were identical to the last year. The entire 24,483 is produced by the decision to weight nearer years more heavily. Change the rate and the number changes; set the rate to zero and the number vanishes.

So an argument that faster AI is worth it on health grounds, built on the benefit arriving sooner, is an argument standing on the discount rate. That is a

respectable place to stand — but it should be said out loud, because readers hear "the medicine arrives two years earlier" as a claim about medicine and not as a claim about discounting.

The paper proves a companion point that runs against the comfortable reading. Under discounting, the gain from making a candidate arrive sooner dominates the gain from raising the probability that it reaches patients at all — not the reverse. If your intuition said the probability channel was the valuable one, the algebra disagrees, and it disagrees in a way that makes the case for speed rest even more heavily on the discount rate.

Two results that refuse to settle each other

This is the newest result in the paper and the one I was most tempted to overstate.

There are two different questions you can ask about the same model. First: does going faster raise or lower misuse loss? Second: does a laboratory that shortens the two capability distances raise or lower it, at a fixed speed?

It is natural to assume that answering one answers the other. Speed and distance are the same kind of thing, after all — one is the other divided by time. They are not the same question, and the paper proves it twice over.

Under the speed reading the independence is immediate. The quantity that decides the speed question is the hazard's elasticity — how sharply danger responds to the rate of advance. That quantity does not appear in the rule for the facility question at all. So you can move the elasticity across its whole range, watch the speed answer flip, and the facility answer does not budge.

Under the level reading it takes a counterexample, and the paper supplies one with numbers. Hold everything fixed except the distance the countermeasure waits on. As that distance crosses 0.833, the speed answer changes sign: accumulated danger responds at +0.747 when the distance is 0.6, exactly 0.000 at the boundary, and −1.813 when the distance is 1.4. A full reversal. Meanwhile, a laboratory move that shortens the countermeasure's distance by 0.05 lowers

misuse loss by 0.080 at every one of those points. One sign swings from positive through zero to negative; the other sits still.

And the counterexample runs the other way too. At one point in that range, two laboratory moves of identical size go in opposite directions: shortening the countermeasure's distance lowers loss by 0.080 while shortening the misuse distance by the same 0.05 *raises* it by 0.067 — which is convention A from Part 3 doing exactly what Part 3 said it does.

So the two questions must be asked separately. An argument that establishes one and asserts the other has skipped a step, and the step does not come free.

And one that is the same only in form

The paper originally made more of this next one than it should have, and the correction is worth seeing because it is the kind of mistake that is easy to make and hard to notice.

Shortening the distance a countermeasure waits on lowers misuse loss, and the expression for how much has the same *shape* under every one of the three choices. True. For a while the paper presented that as a headline: a result invariant to everything the paper could not settle.

It does survive everything. But the sameness is structural rather than surprising, and the reason is almost disappointing once you see it. That distance enters the problem at exactly one point — the date the countermeasure arrives — while each of the three choices governs something else. The danger reading sets how high the plateau is. The exposure reading sets where counting begins, which that distance does not move. The convention sets how the profile responds to the *misuse* distance, which this calculation holds fixed. Three choices, three different objects, none of them the one point where the distance acts. Written out as a shift in time, the algebra is elementary.

And the sameness is of the form, not the magnitude. At one illustrative parameter set the marginal value is 0.750 under convention A, 0.250 under convention B, and

0.715 under the speed reading. Same expression, three different numbers, differing by a factor of three — because each case substitutes its own plateau hazard. A reader told that a result is “invariant across all three choices” could easily hear that the quantity is stable. It is not; the formula is.

What is genuinely this paper’s contribution is the thing underneath — splitting the deployment wait into the part capability sets and the part adoption sets, and the floor that follows from the split. The invariance is a consequence of that, not a discovery on top of it. The paper now says so, and says it below the conditional results rather than above them.

The ledger rule

The sixth survivor is the one from Part 2: every year of healthy life gained or lost is counted in exactly one place. It holds under every choice because it is true by construction.

It belongs on the list, and a reader scanning for results should know what kind of thing it is. It is not a finding about the world. It is a promise about the arithmetic — and like the invariance above, it would be true in a world where every number was different.

And the negatives, which also hold throughout

Three things fail to follow no matter which way the choices go, and they are as much a part of the paper’s output as the positives. A paper that reported only what it established would misrepresent what it did.

No overall sign for the whole balance follows, anywhere, under any branch. No ranking of the available interventions follows. And the paper gives no decision rule for acting while the balance is unidentified — declined explicitly, in its own section, rather than left for a reader to discover by its absence. Part 9 argues that the refusal is the honest answer rather than a dodge.

Part 5 — What depends on the choice, and on which one

Part 4 was the survivors. This part is everything else, and it is the larger pile.

The discipline here is the paper's main contribution, so it is worth saying exactly what it is. Every conditional result below is labelled with the choice it depends on. Not "this result is uncertain" — that would be useless. Instead: *this result exists under the level reading and not under the speed reading*, or *this holds on the ramp-on reading and the arrival-on reading destroys three quarters of it*. A reader can then check whether the branch a result needs is the branch they believe in, which is a thing a reader can actually do.

Truncation: the result that needs two conditions, not one

Truncation is the most striking conditional result in the paper, and the easiest to misreport.

The mechanism: suppose the dangerous capability is far enough away that, at the current rate of progress, it would not be reached inside the window you are counting over. Then it contributes nothing to the ledger — not because it is safe, but because it is outside the frame. Now accelerate. The capability is reached sooner, and at some point it crosses back inside the window and starts being counted.

So acceleration created a cost out of nothing but its own speed. No parameter of the danger changed. The hazard profile is what it was. The only thing that happened is that the window now contains an event it did not contain before.

With numbers: put the dangerous capability ten units away and progress at 0.3 units a year. It arrives in year 33, past a thirty-year window, and the plateau is never counted. Raise progress to 0.4 units a year and it arrives in year 25 — five years of plateau hazard now fall inside the window, and at a plateau hazard of 1

that is five full units of accumulated danger that did not exist in the ledger before the acceleration.

Two conditions, and both matter.

First, it is level-set only. The mechanism is about *arriving* at a dangerous level sooner. The speed reading has no level to arrive at — danger there is a function of the rate of advance, which has no threshold to cross. Under the speed reading truncation is not a smaller effect. There is no such effect.

Second, it only operates while the capability lies beyond the horizon's reach. Once the capability is already inside the window at the current speed, there is nothing left to pull in, and acceleration goes back to the plateau story of Part 4. Truncation is a boundary phenomenon.

Anyone citing truncation as a reason to worry about acceleration needs both conditions, and the second one is quantitative: it is a claim that the dangerous capability is further away than thirty-odd years of current progress. That is a strong claim and the paper does not make it.

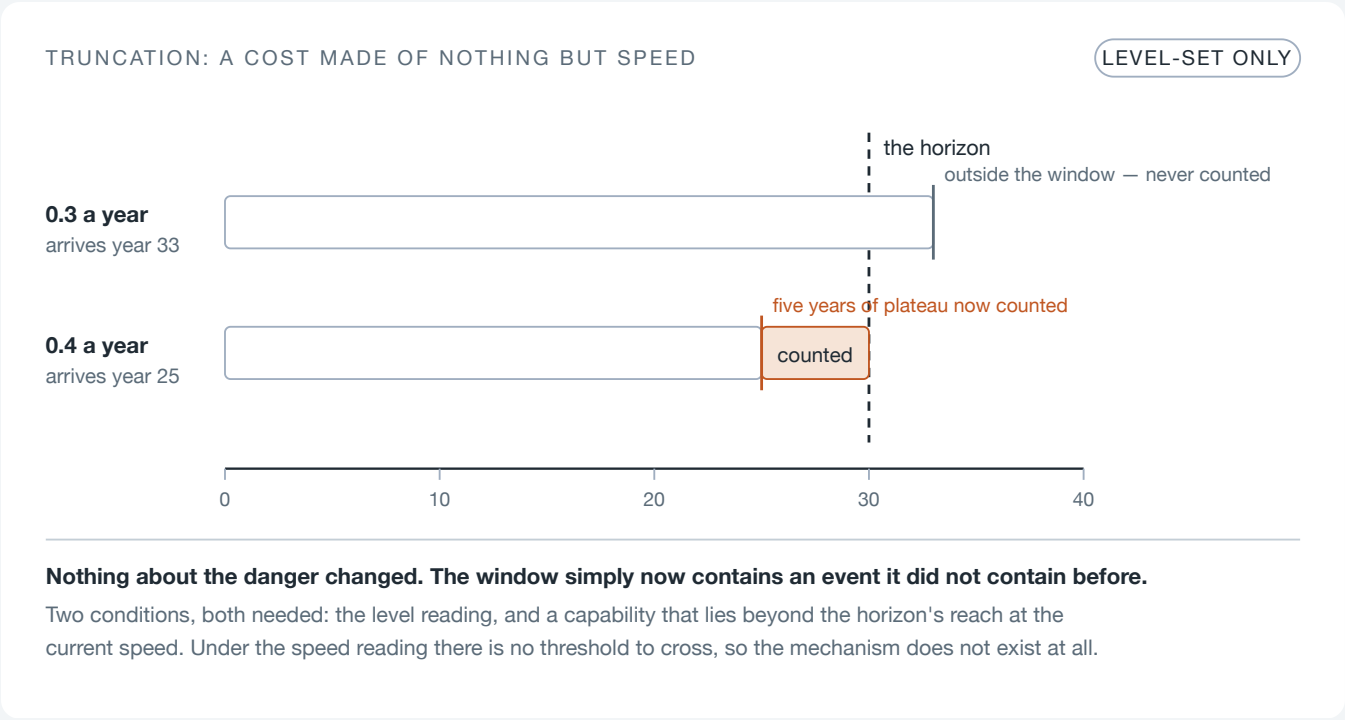


Figure 14. Truncation needs the level reading and a capability beyond the horizon's reach. It is a boundary phenomenon, and the second condition is a

strong quantitative claim the paper does not make. Illustrative values.

The defence clock has two different shapes

The paper's central conditional result is what it calls the defence clock: a race between the danger arriving and the countermeasure arriving, both of them moved by the same speed.

The intuition is clean. Acceleration brings the dangerous capability closer, which is bad, and brings the countermeasure closer, which is good. Which wins depends on how much of each one's wait is capability-set. The reason it is worth a section rather than a sentence is that the race has *two entirely different forms* depending on the first choice, and they are not two expressions of one idea.

Under the level reading, it is a race between two distances. Acceleration lowers misuse loss when the countermeasure's required distance, weighted by how much the countermeasure actually buys — the drop from plateau hazard to residual hazard — exceeds the misuse distance weighted by how much of the ramp is above the ramp's own average. Both sides are *distances*, multiplied by hazard gaps. The quantities a policymaker would need are two capability distances and two hazard levels.

Under the speed reading, it is a race between two ratios. No distances appear. The test is whether the hazard's elasticity — the percentage rise in danger for a percentage rise in speed — is smaller than the capability-set share of the countermeasure's wait, scaled by how much of the hazard the countermeasure removes. The quantities a policymaker would need are an elasticity and a share.

These are different empirical programmes. Under one, you fund work to estimate how far away two capabilities are. Under the other, distances are irrelevant and you fund work to estimate an elasticity. The choice between the branches is not a technicality; it determines what would even count as the relevant evidence. And since nothing in the literature chooses the branch, nothing in the literature tells you which measurement programme to run.

A third variant: on the arrival-on reading, the *gap* between the two distances enters rather than the distances separately. Three readings of one mechanism, three different things to measure.

Four disagreements, and what they are not

The two danger readings sometimes disagree about whether speed raises or lowers misuse loss. The paper found four such cases and states them carefully, because this is the easiest place in the whole paper to overclaim.

Two qualifications come attached, and they are load-bearing.

All four require the ramp-on reading. They are not a general feature of the model. They are what happens when you count danger from the beginning of the capability path. That is one of the two readings, carried co-equally, and it is not the paper's preferred one — the paper has no preferred one.

Opposite signs on the misuse term do not mean opposite rankings of speed. This is the one I would most like a reader to take away. The misuse term is one term in a balance that also contains the benefit, the facility channel and the hospital channel. Two specifications can disagree about the sign of one term and still agree about whether faster is better overall, if the other terms dominate. "The two readings disagree about misuse" and "the two readings disagree about speed" are different statements, and the paper proves only the first.

So the four disagreements are a result about the fragility of one term, not a result about the fragility of the conclusion. A reader who comes away thinking the paper shows the whole question flips depending on an unsettleable choice has read it as more dramatic than it is.

Compared like with like, only one disagreement survives

Here is the part that changed my own view of the four disagreements, and it is a methodological point worth more than the result.

The four cases were found by comparing the two danger readings on the ramp-on exposure reading. But the exposure reading is a separate choice, and it is possible to compare the two danger readings consistently on the *arrival-on* reading instead.

Do that, and three of the four disagreements vanish. One survives.

This is what it looks like when a comparison is doing work that the thing being compared is not. Three of the four were not disagreements between the two descriptions of danger; they were artefacts of counting exposure from the start of the path, showing up as disagreement because the two descriptions respond differently to the length of a period that one of them arguably should not have been counting at all.

The paper reports both: four on the ramp-on reading, one when compared like with like on the arrival-on reading. It would have been easy to report only the four — it is the more striking number, and it was in an earlier draft on its own. Reporting both is the honest version, and the one survivor is more interesting than the four, because it is the one that is actually about the disagreement.

There is a second place the choice bites, and it is not about misuse at all. The choice of danger reading moves a threshold on the *health benefit* itself, whenever the loss per misuse event is anything above zero. The paper states this under a specific normalisation — a rule fixing one reading's hazard against the other's, so that the two are comparable in the first place — and the normalisation is part of the result, not a convenience behind it. Without it the comparison is between quantities on different scales, which is how one gets apparent disagreements that are only unit mismatches.

The same words, opposite effects

Part 3 showed the two conventions with numbers: halve the distance to the dangerous threshold, and under convention A every point on the path becomes

more dangerous while the summit stays as high, while under convention B the path is untouched and the summit drops to a quarter of its height.

The result that follows is the one I find most useful to hold in mind when reading anybody's argument about laboratory policy, including my own.

Under the level reading, on the ramp-on exposure reading, and under conditions the paper states precisely, a facility's effect through the misuse distance **flips sign between the two conventions**. The same intervention, described in the same English, raises misuse loss under one faithful reading of the description and lowers it under the other.

Two things follow, and they point in different directions.

The first is a warning about language. "This work brings dangerous capability closer" is not a specification of an intervention. It is a sentence with at least two models behind it that disagree about the sign of its effect. Any argument built on that sentence has to say which it means, and most arguments — in the literature and in conversation — do not, because until you write the model down it does not occur to you that there is a choice to make.

The second is narrower and more reassuring. Under the speed reading, the misuse distance carries zero weight in the facility comparison on the ramp-on reading, which means the whole dispute between the conventions simply does not arise there. The conventions are a level-set problem. If you believe danger is set by the rate of advance rather than the level reached, you are spared this particular ambiguity — and you have given up truncation and the two-distance form of the defence clock to get there.

Which is the shape of the whole part. Every branch buys clarity somewhere and pays for it somewhere else, and the paper's job is to show the receipt rather than to pick the branch.

A second kind of not-knowing, and a bound that is only half built

Everything above is one kind of ignorance: we do not know which model to write down. There is a second kind underneath it, and the paper separates them because they call for different responses.

Suppose you settle the choices. Pick the level reading, pick an exposure convention, pick a facility convention, and take the resulting signs as given. You still do not know the numbers — the health flow from the benefit, the loss per misuse event, the probability a candidate succeeds, the conversion from a hospital death into healthy years. So the question becomes: could some admissible assignment of those unknown numbers flip the sign of the whole balance between two speeds?

That is a well-posed question and the paper does not fully answer it. It says so, which is the part I want to draw attention to.

Answering it in full requires a construction with several components. **One component is done:** a bound on the hospital channel, and the margin condition that follows from it. The rest is not, and the paper sets out what remains rather than implying the job is finished.

The piece that is done is instructive about the whole enterprise. The bound covers one component of the hospital term — the one the causal design actually measures, direct in-hospital mortality among patients already admitted when an attack began. Excess deaths per attack come out at 0.568 to 0.905, which is the authors' own count of 42 to 67 divided by the 74 attacks in the sample. That is the paper's arithmetic, graded WEAK, and it is a range across reported estimates rather than a confidence interval — two different things that look alike on a page.

Then the bound stops, because no conversion from an excess death into healthy years exists. So the paper does the only honest thing available: it states the bound **as a function of the conversion**, evaluated at three anchors rather than at one. Not a number with error bars. A number conditional on a quantity nobody has measured, reported at three values of that quantity so a reader can see how much it matters. Part 8's factor of 260 is how much it matters.

Two further details, both about method. The bound was computed twice, independently and blind, from the same inputs — two separate calculations that did not see each other, which is a weak form of replication but better than none. And only the working-paper version of the hospital study enters it, because the two versions are never mixed and the published full text was not read.

So there are two layers of ignorance stacked here. Which model, and which numbers. The paper resolves neither, maps the first thoroughly, and maps the second partially while saying where it stopped. An unfinished construction reported as unfinished is worth more than a finished one whose joins are hidden, and I would rather a reader finished it than believed it was already done.

Part 6 — The defence clock, taken apart

One result in the paper does more work than the others, because it is the one a policy argument would actually lean on. If somebody tells you that accelerating AI is good for health because the countermeasures arrive sooner too, this is the result they are reaching for, whether or not they know it exists.

It deserves a part to itself, at a pace slower than the rest of the guide.

Two clocks, and only one of them obeys speed

Two things are racing, and each has its own clock.

The first clock runs to the dangerous capability. Its length is the distance to that capability divided by the speed of advance, and that is the whole of it. Double the speed and the clock halves.

The second clock runs to the countermeasure, and it has two segments. There is the part that waits on capability — somebody has to be able to build the thing — which again is a distance divided by the speed. And there is the part that waits on everything else: trials, approval, manufacture, distribution, uptake. Call that the adoption segment.

Speed compresses the capability segments of both clocks. It does nothing whatever to the adoption segment, which runs on calendar time and on institutions.

That asymmetry is the entire mechanism. Acceleration shortens the run-up to danger by the full proportion, and shortens the run-up to the countermeasure by less than the full proportion, because part of that wait does not respond. So the gap between the two — the exposed period, when the danger exists and the countermeasure does not — can either shrink or grow, and which one it does is an arithmetic question with a definite answer once you say what the pieces are.

Under the level reading: a race between two distances

Take the level reading first, where danger depends on how far capability has got.

The exposed period runs from the moment the dangerous capability is reached to the moment the countermeasure lands. During it, the hazard sits on its plateau. After the countermeasure lands, the hazard drops to some residual level — not to zero, because no countermeasure is complete.

Acceleration does two things to the accumulated danger, and they pull opposite ways.

It brings the countermeasure forward, which ends the exposed period sooner. The value of that is the countermeasure's capability distance multiplied by how much the countermeasure actually buys: the drop from the plateau hazard down to the residual.

It also brings the dangerous capability forward, which starts the exposed period sooner and replaces time that had been spent on the ramp — where the hazard was still climbing — with time on the plateau. The cost of that is the misuse distance multiplied by the gap between the plateau hazard and the ramp's own average.

Acceleration lowers misuse loss when the first product exceeds the second. Two distances, two hazard gaps, and the comparison between them is the result.

Where the line falls, and what sits on each side

Numbers make this one tractable, and the paper's own illustrative set is as good as any.

Let the hazard climb as the square of progress along the path, reaching 1 at the threshold — so its average along the ramp is one third. Let the residual after the countermeasure lands be 0.2. Put the misuse distance at 1 and the speed at half a unit per year, so the dangerous capability arrives in year 2. Give the countermeasure an adoption segment of one year.

The countermeasure buys a drop of $1 - 0.2 = 0.8$. The misuse side costs $1 - \frac{1}{3} = 0.67$, carried by a distance of 1. So acceleration helps when the countermeasure's distance, times 0.8, exceeds 0.67 — that is, when its distance exceeds **0.833**.

Now stand on either side of that line.

With the countermeasure's distance at 0.6 — below the line — acceleration *raises* accumulated danger, at a rate of +0.747. The countermeasure was close enough that speeding up did not buy much on that side, while it bought a good deal of extra plateau on the other.

With the distance at 1.4 — above the line — acceleration *lowers* it, at −1.813. The countermeasure was far enough away that pulling it forward was worth more than the plateau it cost.

At exactly 0.833, acceleration does nothing at all.

So the same model, with nothing changed but how far away the countermeasure is, says accelerate, says it makes no difference, and says do not accelerate. That is not a defect. It is the result: the sign is determined by a comparison of two

quantities, and anybody asserting the sign without those quantities is asserting something the model does not give them.

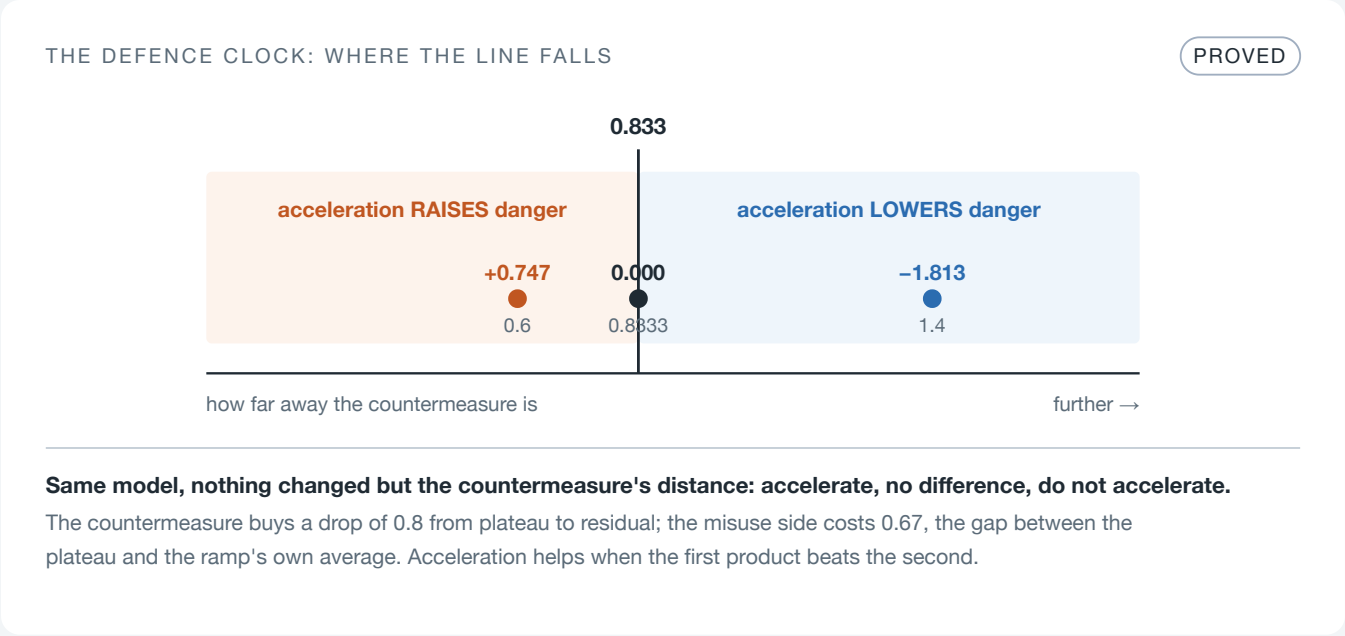


Figure 15. The sign is settled by a comparison of two quantities, and anyone asserting it without them is asserting something the model does not supply. The paper's own illustrative parameter set.

Under the speed reading: a race between two ratios

Switch to the speed reading and the shape of the answer changes completely.

Here danger depends on the rate of advance, not on the level reached. There is no plateau to arrive at and no ramp average to compare against. The distances drop out of the comparison entirely.

What replaces them is a pair of ratios. The first is the hazard’s elasticity: if the speed of advance rises by one per cent, by what percentage does the hazard rise? The second is the share of the countermeasure’s wait that capability sets, scaled by the fraction of the hazard the countermeasure removes.

Acceleration lowers misuse loss when the elasticity is below that scaled share.

Put numbers in it. Suppose six tenths of the countermeasure’s wait is capability-set, with the rest adoption, and suppose the countermeasure removes eighty per

cent of the hazard when it lands. The threshold is $0.6 \times 0.8 = \mathbf{0.48}$. If a one per cent rise in the speed of advance raises the hazard by less than about half a per cent, accelerate. If it raises it by more, do not.

Notice what a policymaker would need to know under each reading. Under the level reading: two capability distances and two hazard levels. Under the speed reading: an elasticity and a share. Not one quantity in common. These are two different research programmes, and nothing in the literature chooses between them — which means nothing in the literature tells you which programme to fund.

On the third reading, where the clock starts at arrival rather than at the beginning of the path, it is the *gap* between the two distances that enters rather than the distances themselves. Three readings of one mechanism, three different things to go and measure.

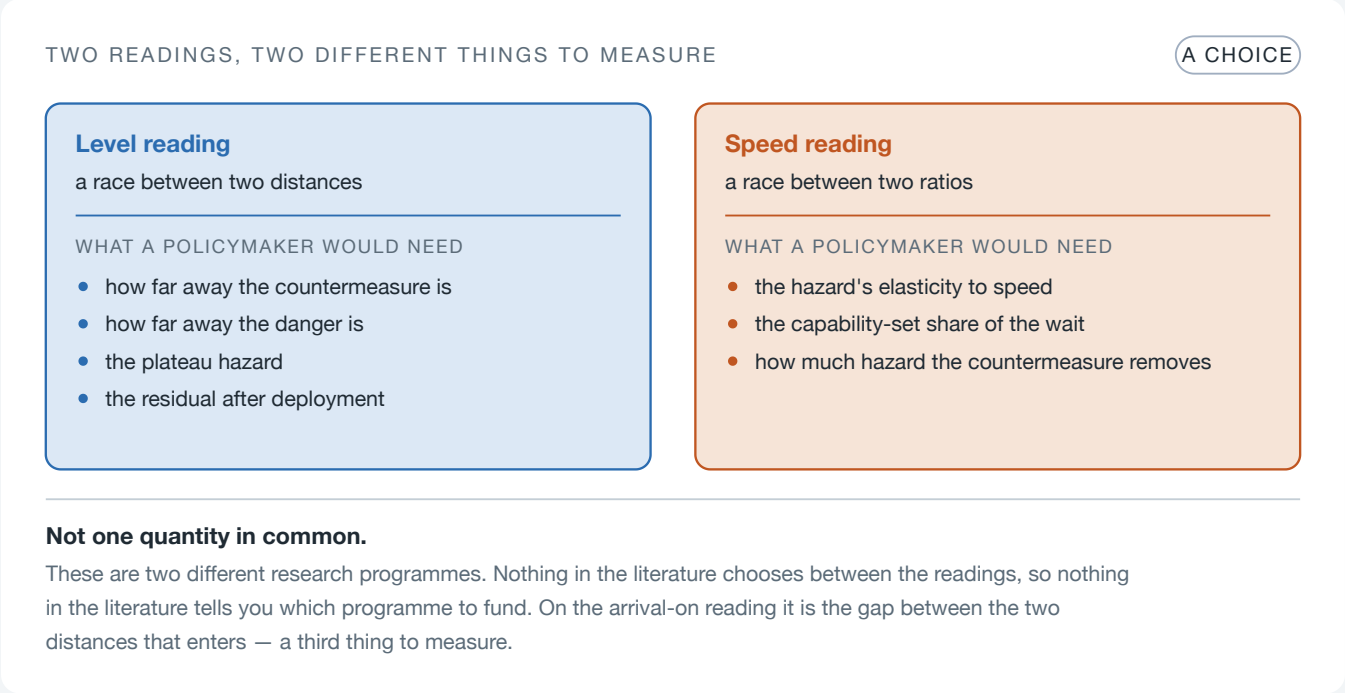


Figure 16. The choice is not a technicality: it decides what would even count as the relevant evidence. A drawing of the choice, not of data.

The corner the paper does not sign

Two regions of this picture are not covered by the result, and both matter.

The first is the adoption floor from Part 4. Once the countermeasure's wait is set by adoption rather than capability, the capability-set share goes to zero and so does the whole benefit of acceleration on that side. In that region both readings agree that speed hurts, which is a rare moment of consensus in this paper and worth noticing: where the countermeasure is stuck in approval, the two descriptions of danger stop disagreeing.

The second is the corner where the countermeasure arrives *before* the dangerous capability. That is a different regime — there is no exposed period to shorten — and the paper does not sign it. It says so rather than extending the inequality past where it was derived, which is the kind of restraint that is invisible when it is done and embarrassing when it is not.

When both mechanisms are present at once

A reasonable objection to the whole arrangement: why insist on choosing between level and speed? Surely the real hazard has some of both.

The paper takes the objection seriously enough to work it out, and the answer is not comfortable.

If the countermeasure's timing is mixed but the hazard is purely speed-set, the test is a clean share comparison, and it reproduces the earlier paper's result exactly.

If the *hazard* is genuinely mixed — part level, part speed — a share test of that kind survives only under conditions, and the condition it needs is a closing clock whose post-capability segment does not lengthen as speed rises. Where the hazard is additive in its two components, the test becomes a comparison between the speed component's share of the loss, weighted by its elasticity, and the capability-set share of the loss. At one extreme it collapses to the earlier paper's test; at the other to the level-set result. That is tidy, and it is the best news in this subsection.

The bad news is the plateau. On the plateau, with a mixed hazard, **no share condition exists at all**, on either exposure reading. Under a fixed horizon with a non-decreasing level component, the threshold is non-positive — which means acceleration raises loss, with no comparison left to make.

So mixing the mechanisms does not rescue the comfortable conclusion. It preserves it in one corner and removes it entirely in the corner where the dangerous capability already exists.

Why this is the result a policy argument would lean on

Stand back from the algebra and the practical content is this.

"Acceleration brings the cure as well as the disease" is not wrong. It is a statement whose truth depends on four things: how far away the countermeasure is, how far away the danger is, how much of the countermeasure's wait is institutional rather than technical, and how much of the hazard the countermeasure actually removes. Change any one of them and the sign can change.

Three of those four have never been measured for any biological scenario. The fourth — the institutional share of a countermeasure's wait — is the one we could estimate today from vaccine and drug approval histories, and Part 10 puts it at the top of the list for exactly that reason.

Which leaves the defence clock as a result that is genuinely useful and genuinely unusable: it tells you precisely what you would need to know, and the knowing is not available.

Part 7 — What the evidence reaches, and where it stops

Everything so far has been structure: a unit, three choices, what survives them and what does not. This part leaves the mathematics and asks the question anybody sensible asks first. *What do we actually know?*

The answer is less than the public conversation implies, and the shortfall is not where most people expect it. It is not that the evidence is thin. On the one thing that has been measured properly, the evidence is strong. It is that the measured thing sits three steps away from the thing the argument needs, and the three remaining steps have no measurements at all.

Five links, and the evidence reaches the third

Between a capable model and a health loss there are five separate claims, and each is a claim in its own right.

1. **Access.** Someone has the model, or something like it.
2. **Intent.** Someone wants to cause harm.
3. **Task performance.** The model helps with the relevant tasks.
4. **Realised misuse.** An attempt actually happens.
5. **Consequences.** The attempt causes harm, in some amount.

The paper's question — how many healthy years are at stake — lives at the fifth link. It needs a probability of events and a loss per event.

The evidence reaches the third and stops. That is the whole finding of this part, and the rest is detail about how it stops.

The reason this matters is that a grade at one link does not transfer to the next. Strong evidence about task performance is not weak evidence about consequences; it is *no* evidence about consequences, in the same way that a strong measurement of somebody's running speed is no measurement of whether they will run. People slide across the links without noticing, and the slide is invisible because each step sounds reasonable on its own. The model is very good at the test; the test is about a dangerous field; therefore the danger has risen by — and there the sentence has to stop, because nothing measured continues it.



Figure 17. Each link is its own claim, and the slide across them is invisible because every step sounds reasonable alone. Graded from the paper's evidence tables.

The benchmark: strong, and about the wrong thing

On a purpose-built test of virology knowledge, a publicly available model outperformed 94 per cent of expert virologists working within their own sub-areas.

That is a real result, and the paper grades it STRONG for what it is. A hard examination, a human expert baseline, a careful two-stage review of the items. There is no methodological complaint to make.

It is also evidence about an examination. It supports no estimate whatever of how likely an attack is, how often one would succeed, or what it would cost in health. It is the third link, measured well.

The honest way to state what it establishes: a widely available system now exceeds most domain experts on a written test of knowledge in a field where some knowledge is dangerous. That is worth knowing and it is not nothing. It is also not a quantity that can enter any term of the paper’s ledger.

The one controlled trial, in detail

One study in the literature does the thing an empirical scientist would ask for: a randomised comparison with a no-AI arm. It is the only such design the paper located, and it is not a developer evaluating its own product.

The design, stated plainly. Twelve cells of three researchers each, randomised across four scenario strata — four cells working with internet access only, four with access to one large language model, four with another. Each cell produced one written plan. Each plan was scored on a nine-point viability scale by eight adjudicators, four drawn from biology and four from security, under a structured protocol designed to stop the loudest voice dominating.

The result is a null, and the direction is worth noting. Cells with model access produced plans scored **0.22 points lower** in viability than cells with internet access only. The arm means: 3.52 for internet only, 3.30 with a model — 3.64 for one of the two models, 2.96 for the other. The t-statistic is −0.49 and the p-value 0.64.

Two features of the scores matter more than the means.

Every plan in every arm, the control arm included, scored somewhere between the rubric's labels for "untenable" and "problematic". No arm produced anything the adjudicators thought would work. That is a floor effect in the control arm as much as in the treatment arms, and a comparison in which nobody succeeds cannot tell you much about what would help somebody succeed.

And the report prints estimates, t-statistics and p-values, but no confidence interval, no standard error, and no inter-rater statistic anywhere — and its significance tables exclude three cells that were not randomised.

The paper grades the study MODERATE. Design transparency raises it above weak; twelve cells, the missing interval, and the variance problem below keep it below strong.

What the trial could have detected

Here is the load-bearing number in the whole evidence base, and it is the authors' own.

Their design, they say, could have detected a change of about **1.0 point** on that nine-point scale. The difference they observed was 0.22. So the study was built to notice an effect roughly five times larger than the one it found.

A study that could not have seen the effect it estimated has not measured that effect. It has reported a number whose main property is that the design could not distinguish it from a range of other numbers.

And the authors attach a qualification to their own sensitivity figure that cuts deeper. The calculation assumes everything else held constant — and their report says elsewhere that *cell composition was likely a much greater source of variability than model access*. Three researchers differ from three other researchers more than one tool differs from another. If that is right, the design was straining to hear a whisper in a room where the furniture was being moved.

The authors state their own limits and decline to extend the result: adjudicated viability of a written plan, one rubric, the planning phase only, the guardrailed models of mid-2023, none of them fine-tuned. Their sentence about it is the one I would put on the wall of any room where this gets discussed. Their results, they write, "do not and could not disprove the existence of risk."

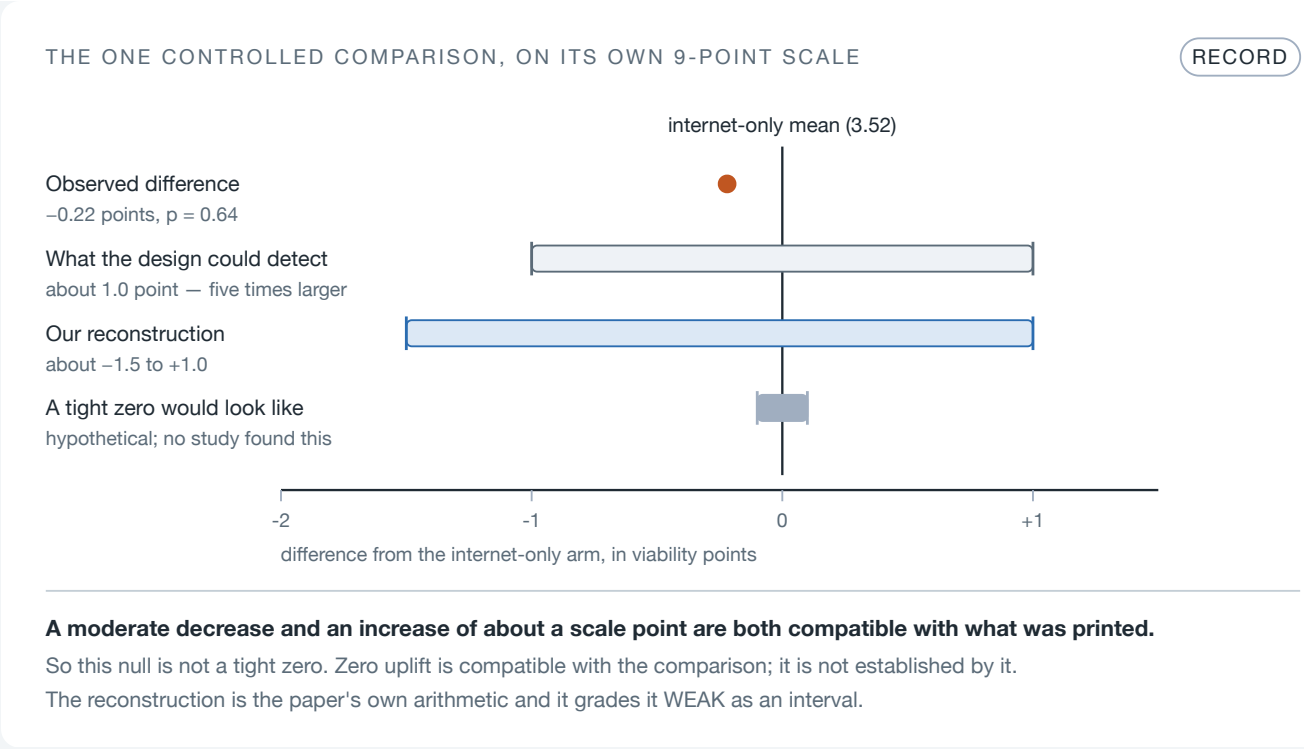


Figure 18. A minimum detectable effect is a property of a design under an assumed power, not a statement that smaller effects are absent. Published statistics and the paper's reconstruction; drawn to scale.

Our own reconstruction, and why it is weak

Because no interval is printed, the paper reconstructs one — and then spends more words on why the reconstruction is weak than on the reconstruction.

The arithmetic is simple. A difference of 0.22 and a t-statistic of 0.49 imply a standard error of about 0.45. Carry that through and you get an approximate ninety-five per cent envelope running from about **−1.5 to +1.0** viability points.

The honest caveats, all of which the paper prints.

Both printed inputs are rounded to the nearest hundredth, so the implied standard error is itself only pinned to about ±0.02. The inversion needs five named assumptions, including an ordinary two-sample test on twelve cell means and a range for degrees of freedom that the report never states — searched from 3 to 10, with compatibility against the printed statistics binding at about 4.27. Two people did the reconstruction separately; neither could recover a unique interval from

what is printed, and the paper carries the wider of the two envelopes they found rather than the one that suited it better.

Most important: the real design was stratified, and the inversion assumes it was not. The envelope is therefore for a comparison the authors did not run.

So the paper grades its own reconstruction **WEAK as an interval** — and MODERATE only for the one qualitative point it supports, which is that the data do not pin the effect anywhere near zero. It is explicitly not allowed to constrain any parameter of the model. It is carried as a caution and nothing else.

I have gone through this in detail because the reconstruction is the most criticisable thing in the paper, and because the right response to that was to print the criticism rather than to drop the arithmetic or dress it up.

An insensitive null is not a tight zero

This is the error the paper most wants a reader to stop making, and it runs the opposite way from the error people expect.

A minimum detectable effect is a property of a study's design under an assumed level of power. It is not a statement that smaller effects are absent, or impossible, or unlikely. A study powered to detect one point on a nine-point scale, finding 0.22, has established nothing about whether the true effect is 0.22, or zero, or 0.8 in either direction.

On the reconstruction above, a moderate decrease and an increase of about a full scale point are both compatible with what was printed. Neither direction is licensed.

So: zero uplift is compatible with this comparison. It is not established by it. The study constrains the misuse hazard in neither direction and constrains the loss per event not at all.

The tempting error is to read the null as evidence of safety. The study's own authors do not make it, and the paper is careful not to make the mirror-image error either — the null is not evidence of danger being hidden, it is simply not evidence about the size of the danger.

Narrowing this would need a design very much larger than twelve cells. The paper located none, and notes that the expert-consensus review it read says the field has no standardised benchmarks for uplift at all, with real-world impact unclear because of secrecy, testing prohibitions and immature evaluation methods. That is a search result rather than a proof of non-existence, and the gap is contingent on those three conditions rather than inevitable.

One report of attempted misuse, and what it does not supply

One record of attempted misuse was located, and the paper's treatment of it is a good illustration of grading.

In September 2026 a model developer reported that it had disrupted several potential plots that year involving scientists whose research with its models could have helped develop biological weapons. The company said it could not determine whether the research served a legitimate or a nefarious purpose, and that it erred on the side of caution given that uncertainty — which is a creditable thing to say and worth crediting.

Three limits travel with it, and the paper states all three.

First, provenance. The paper read a press account of the report, reaching it by syndication, with one paragraph of that account unread; the company's own report was not located. A claim read at two removes is not a claim read.

Second, structure. The figure is a count of plots the company says it detected and chose to report. There is no denominator of attempts, and no outcome for any of them. A count of detections with no denominator can rise because attempts rose, or because detection improved, or because reporting policy changed.

Third, position. It speaks to the fourth link — that attempts occur — and not to the fifth. Evidence that someone tried is not evidence that harm followed.

The paper grades it WEAK and records that it moves none of the three misuse quantities. It is also the single strongest available indication that the fourth link is not empty, which is why it is reported at all rather than set aside.

The two strongest pieces do not even overlap in time

One more fact deserves its own heading, because it is easy to miss and hard to unsee.

The benchmark result concerns a recent, publicly available model. The randomised trial concerns the guardrailed models of mid-2023, none fine-tuned.

So the strongest evidence that models are capable, and the only controlled evidence about whether that capability helps anybody do anything, are about **model generations that do not overlap in time**. They are not two measurements of one question. They are measurements of two different questions about two different technologies, separated by several model generations in a field where a generation is roughly a year.

Anyone combining them into a single picture is doing something the data does not support, and the direction of the error is not obvious — the newer models are more capable, but the trial's null tells us nothing about them either way.

Who wrote what, on both sides

The paper discloses the interests in this literature rather than grading around them, and the disclosure runs both ways.

The benchmark's authors are at biosecurity organisations, which is to say at institutions whose purpose is to identify this class of risk. The review that finds no recorded real-world harm is led from one AI company and has a co-author at another — two model developers among otherwise academic co-authors — and its

argument is that the available literature is methodologically immature and not transparent enough to settle the question. The report of disrupted attempts is a vendor's account of its own product.

None of this bears on the design of any of the three, and none of it is a reason to set any of them aside. The benchmark is well built whoever built it.

All of it is a reason not to read agreement inside one camp as independent corroboration. Two findings pointing the same way, from institutions with the same interest in the answer, are closer to one finding than to two — and the same applies in both directions, which is why the paper discloses the risk-asserting side's interests as carefully as the developers'.

What sits past the third link

To close the part with the plainest possible statement of the gap.

No source the paper read estimates the change in attack probability per unit of capability. None estimates casualties. None estimates healthy years lost. No misuse causing real-world harm is recorded in anything it read. And the expert consensus it read is that the field has no standard for the quantity in the first place.

That is not a claim that no such estimate exists anywhere. It is a report of what a search found, with the search's limits stated. But it is enough to settle what the paper can do: it may write the misuse terms down, and it may not put numbers in them.

Which is why the paper's conclusion is a list of what would settle what, rather than an answer. That list is Part 10.

Part 8 — The one channel with a number, and what it does not license

Part 7 ended with a gap: the misuse channel has no measured magnitude anywhere in what the paper read. This part is about the one channel that does have one, and about the trap that creates.

Ransomware kills people in hospitals, and we can measure it

Criminals encrypt a hospital's systems and demand payment. The hospital diverts ambulances, reverts to paper, cancels procedures and runs degraded for days or weeks. Patients already in its care get worse care than they would have got.

Unlike almost everything in Part 7, this has been studied with a credible causal design, and the effect is there.

What the study actually did

One study identifies the effect, and it is worth describing because the design is the reason to believe it.

It uses a stacked event study and an admission-level difference-in-differences on Medicare fee-for-service claims: **74 attacks on 163 hospitals between 2016 and 2021**, with standard errors clustered by attack, funded by a health-care foundation rather than by a security vendor. The comparison is between patients admitted to a hospital during an attack and comparable patients admitted when there was no attack, with later-attacked hospitals available as an alternative control group.

One limitation belongs in the same breath as the design. Attacks were identified through news reports and federal filings, so attacks nobody disclosed are missing from the sample. If undisclosed attacks are systematically smaller, the estimates are too large; if hospitals hide the worst ones, too small. Nothing in the record settles which.

The paper grades this record **STRONG on direction and MODERATE on magnitude**. Something real is happening and the size of it is less certain than the fact of it.

Two versions of one study, and why the paper reports both

An awkward detail, reported because hiding it would be worse.

The study exists in a published version and a working-paper version, and the paper could read only one of them in full. The published body was behind a paywall it could not get through, so for that version it holds the abstract alone. The working paper it read completely.

They do not say the same thing.

The published abstract reports hospital volume down 17 to 24 per cent in the week of the attack, and in-hospital mortality up 34 to 38 per cent among patients already admitted. Those are relative effects, and they sound enormous.

The working paper reports the same kind of result in absolute terms, with the baseline attached: in-hospital mortality up **0.77 percentage points on a base of 3.72 per cent** in its main specification, with a p-value of 0.07; and up 1.23 points on a base of 3.38 per cent when later-attacked hospitals are used as controls.

The two versions differ in specification and in magnitude, and the working paper's absolute effects are its own estimates rather than arithmetic conversions of the published percentages. So the paper reports both, says which one it read in full, and does not quietly pick whichever is more convenient. A 34 per cent rise in mortality and a 0.77 point rise on a 3.72 per cent base are not obviously the same claim, and a reader deserves to see both numbers rather than the more alarming one on its own.

From percentage points to people: about one death per attack

Percentage points are hard to feel. The study's authors did their own count in a 2023 account of the working-paper version: **42 to 67 excess deaths** among Medicare patients across the whole 2016–2021 period.

Divide by the 74 attacks in the sample and you get roughly **0.6 to 0.9 excess in-hospital Medicare deaths per attack**. On the order of one death per attack.

That division is the paper's own arithmetic, not the authors', and the paper grades it WEAK as a rate and explains why. A systematic review reports a higher range, 69 to 76, on its reading of the published text — which the paper could not read, so it declines to divide that figure, because it does not know the attack count the published version used.

And either count is a floor rather than an estimate. It is Medicare patients only, not all payers. It omits morbidity entirely — people harmed but not killed. It omits deaths after discharge. And it omits patients displaced to other hospitals, who are counted in neither column.

So: about one excess death per attack, as a lower bound, on an arithmetic the paper calls weak. That is the single most solid number anywhere in this literature, and look at how many qualifications it is wearing.

The null that changes what the deaths mean

Now the finding that makes this part more than a tally, and it is a null.

The working paper finds **no effect on 30-day mortality**, and none on readmissions, and none among patients already admitted to neighbouring hospitals.

Hold the first one next to the main result. In-hospital mortality rises during an attack. Mortality measured at thirty days does not. Those two facts together are consistent with something specific: that the attacks **brought deaths forward** rather than causing deaths that would not otherwise have happened in that period. People who were going to die soon died sooner.

If that reading is right — and it is the paper's reading, not the authors', who do not draw it — then the health cost per excess death is very much smaller than a life-expectancy conversion would give, because what was lost is weeks rather than decades.

The paper is careful about the status of this. It is an inference from a null, the weakest kind of inference available, and a null can also mean a study underpowered at thirty days. But it is the most informative thing in the record about the one conversion the whole channel needs, which is why it earns a heading.

The factor of 260, and where it comes from

Here is where the two readings collide, and the size of the collision is the point.

To put hospital deaths into healthy years, you need to know how many healthy years each death cost. That requires the remaining healthy life expectancy of the people who died — and no source the paper read supplies it.

What is available is the mean age of the study's inpatients: **74.21 years**. Not the mean age of the excess decedents, who were probably older and sicker than the average inpatient; the data does not give their ages.

Now take the two readings and convert, holding the World Health Organization's own valuation fixed throughout so that nothing depends on a choice made here.

On the **averted-death** reading, a death at 74 costs the remaining life expectancy at 74, which is about 21.62 years.

On the **displaced-death** reading — the one the 30-day null points at — a death brought forward during an outage costs something like the thirty days by which it was advanced.

The ratio between those is 21.62 years against 30 days: a factor of about **260**.

Sit with that. Not 260 per cent. A factor of 260. The same deaths, the same count, the same valuation of a healthy year, and the health cost differs by more than two orders of magnitude depending on the answer to a question nobody has asked the data.

And notice what the spread is not. It is not sampling error — the study is well powered for the effect it measures. It is not disagreement about the unit — the valuation is held fixed. It is one unanswered question: were these deaths caused, or advanced?

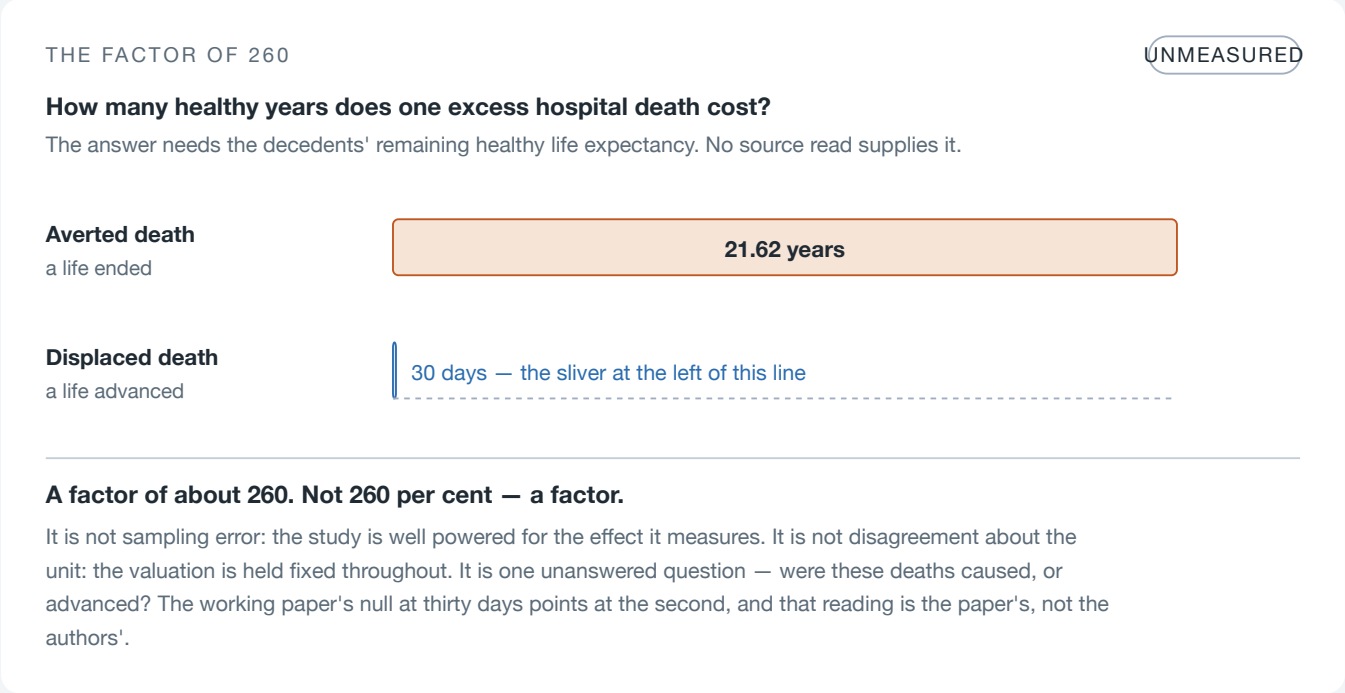


Figure 19. The spread is not the data's and not the unit's. It is the unanswered question of whether these deaths were caused or advanced. The paper's own arithmetic on published figures, drawn to scale.

Displacement, not just damage

A second strand of evidence makes the picture less like destruction and more like diversion.

In all-payer California data, emergency visits fell 16.21 per cent at attacked hospitals in the second week after an attack, and rose by up to 7.10 per cent at nearby hospitals in the third week, with everything recovering inside eight weeks.

The volume result is STRONG. The outcome evidence for whether the displaced patients came to harm rests on a single event with tiny counts and is WEAK. So displacement is the better-supported account of what happens: patients go somewhere else, and whether going somewhere else hurt them is not established.

This cuts both ways for the ledger. Displacement means the harm is smaller than a naive reading of "the hospital stopped working" implies. It also means the harm is somewhere the study was not looking, since the neighbouring-hospital null covers patients already admitted there rather than the ones who arrived because of the attack.

Single-incident studies, with cruder outcomes and shorter outages, report no detectable mortality difference at all. Those can be reconciled with an average effect across 74 attacks in the obvious way: harm is not visible in every attack, and the loss per event is a distribution rather than a constant. The paper says so rather than treating the single-incident nulls as a contradiction.

The channel that delays trials rather than care

One more route, with a number attached and a different shape.

One national attack cut clinical trial referrals by 85 per cent and recruitment by 55 per cent. The evidence is ten survey responses held at abstract level, so it is thin — but the mechanism matters for the model, because it does not reduce the benefit's size. It delays the benefit's arrival.

In the ledger from Part 2 that is a different term, and it has a different consequence: a delay interacts with the horizon, which is exactly the asymmetry Part 2 flagged. Attacks on the systems running trials push the benefit's arrival later, and a benefit that arrives later loses its tail to the edge of the counted window.

There is also evidence that hospital quality measures worsened after data breaches, with the authors proposing remediation work as the cause without observing it. The paper books any such loss once — to remediation or to baseline care, never both, which is the Part 2 rule doing its job.

Measurability and magnitude need not align

Now the point this whole part exists to make.

Attacks on US health-care delivery organisations rose from 43 to 91 a year over the study period, with 44.4 per cent of them disrupting care. No later count was located. So the channel is real, growing in the period measured, and has a credible causal estimate attached.

And here is the thing that should stop a reader cold: **no primary-source report of an AI-enabled attack on a named hospital was located anywhere.**

So the one channel with a measured magnitude is the one channel not empirically connected to the paper's own independent variable. Ransomware against hospitals is measured, and it is not measured as a consequence of AI capability. The misuse channel is connected to AI capability by the whole argument of the field, and it is not measured at all.

Which produces a trap, and it is the reason the paper says this twice. If you form an overall impression from the numbers available, you will conclude that the measurable harms here are modest — roughly one advanced death per attack, possibly worth weeks rather than decades, with displacement rather than destruction. And that impression would be an artefact of *which channel can be measured*, not a finding about where the danger is.

A reassuring reading of what is measured reflects measurability, not safety.

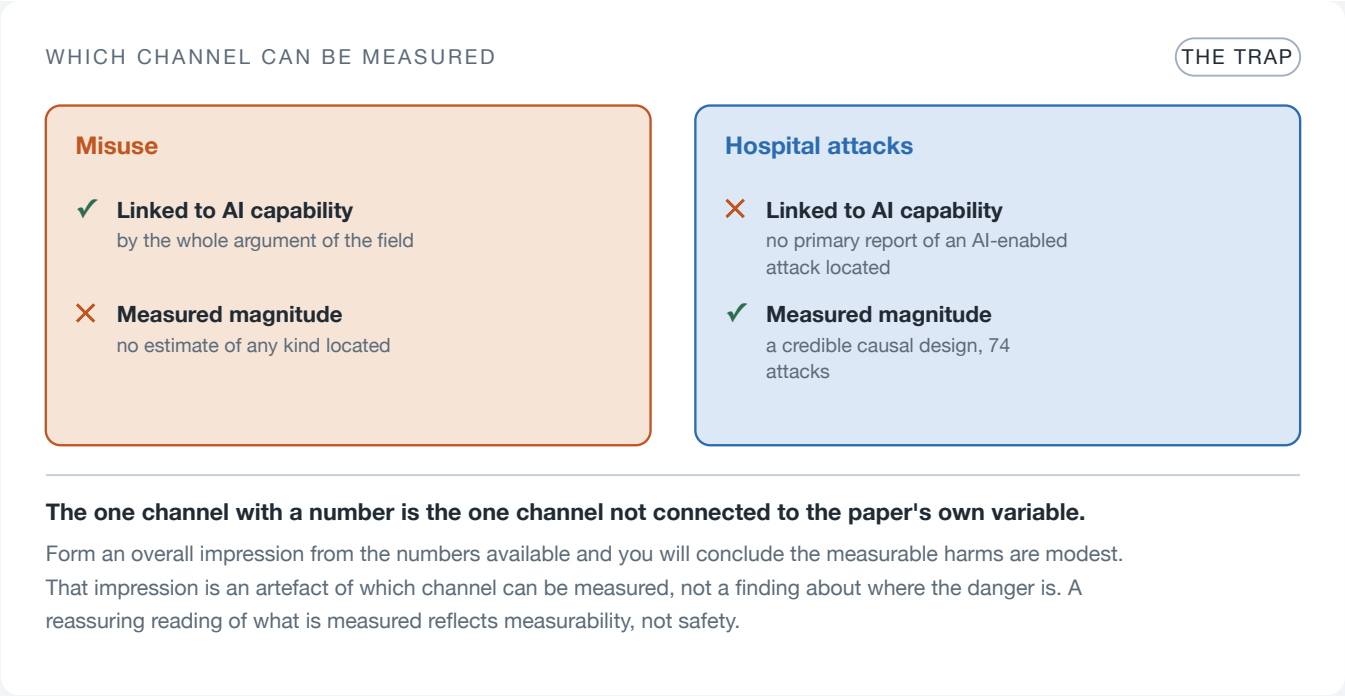


Figure 20. Wherever one harm is frequent, bounded and recorded while another is rare, catastrophic and unrecorded, evidence accumulates on the first and the bias always runs the same way. From the paper's evidence sections.

Why this matters outside this paper

This asymmetry is not special to biological risk, and I think it is the most transferable thing in the paper.

In any field where one harm is frequent, bounded and well recorded while another is rare, catastrophic and unrecorded, evidence accumulates on the first and never on the second. Every honest literature review will therefore be dominated by the measurable channel. Every attempt to “follow the evidence” will weight it. And the direction of the resulting bias is always the same: toward whichever danger happens to be easy to count.

The remedy is not to inflate the unmeasured channel to compensate, which would be inventing a number. It is to say which channel each piece of evidence belongs to, and to refuse to let the balance be settled by the one that happens to have data. That refusal is Part 9.

Part 9 — What the paper refuses to do, and why that is an answer

A paper that spends a hundred pages establishing that the balance cannot be signed is going to be asked an obvious question. *Fine — but what should we do?*

The paper declines to answer, and this part is why. I want to be clear that declining is a position, not an absence. The paper states the refusal in its own section rather than leaving a reader to notice that the recommendation never arrives, because a refusal you have to infer is indistinguishable from an oversight.

The rule you would reach for

Almost everybody, facing a decision where the stakes are large and the probabilities unknown, reaches for the same rule: assume the worse case and act accordingly.

It has a formal name — maximin, choosing the option whose worst outcome is least bad — and an informal one, the precautionary principle. It is the most natural move available, and in many situations it is the right one.

The paper's claim is narrow and specific: **on this structure, that rule cannot be applied.** Not that it is wrong in general, and not that caution is misplaced. That the rule needs inputs this problem does not provide.

What Rawls actually said about it

The best statement of when the rule applies comes from John Rawls, who used maximin to argue for his principles of justice and was careful about its limits (*A Theory of Justice*, 1971, §26).

He gives three conditions.

One: knowledge of the likelihoods is impossible, or at best extremely insecure.

Two: the chooser cares very little for whatever might be gained above the minimum they can be sure of.

Three: the rejected alternatives have outcomes one can hardly accept.

And he is explicit about the rule's status. Maximin is, in his description, a maxim — a rule of thumb that comes into its own in special circumstances — rather than a general guide to choice under uncertainty. The philosopher Stephen Gardiner later adopted the same three conditions as the criteria for a core precautionary principle, and stated plainly that they pick out *sufficient* conditions for precaution rather than necessary ones (Gardiner 2006).

So the question is not whether caution is good. It is whether this problem satisfies the conditions under which the formal rule is defensible. The paper checks them one at a time.

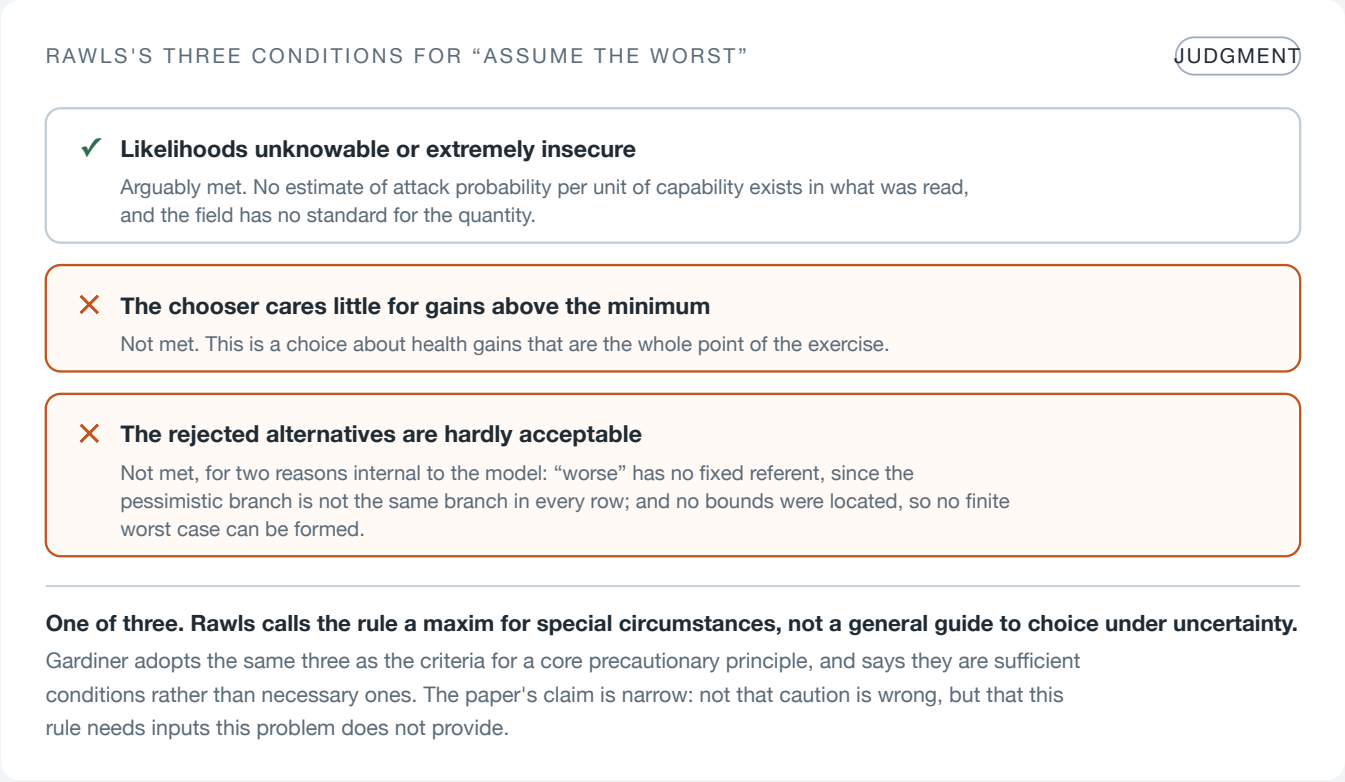


Figure 21. Rawls, *A Theory of Justice* (1971), §26; Gardiner (2006) adopts the same three for a core precautionary principle and calls them sufficient rather than necessary. A judgment about this problem, not a result of the model.

The first condition, which the paper arguably meets

Condition one asks whether the likelihoods are unknowable or extremely insecure.

On the evidence of Part 7, the paper has a decent claim to this. No estimate of attack probability per unit of capability exists in what it read. The expert consensus it read is that the field has no standard for the quantity. “Extremely insecure” is, if anything, generous.

So the first condition is arguably met, and the paper says so. One out of three.

“The worse reading” has no fixed meaning here

Condition three needs a worse case to identify. Here the paper’s own central finding gets in the way, and the obstruction is internal to the model rather than a gap a longer search would close.

Acting as though the worse reading held requires a fixed referent for “worse”. There has to be a branch that is reliably the pessimistic one.

There is not. The signs flip across the rows of the paper’s own non-identification table. In some cells the level reading gives the gloomier answer; in others the speed reading does. Which branch is “worse” depends on which row you are standing in — and that is the paper’s result, not an accident of presentation.

So a precautionary rule here would have to be applied cell by cell, and the consequence of that is worth stating carefully: such a rule **could select an action that is optimal under no single reading**. Take the pessimistic branch in each row independently and you assemble a policy that no coherent view of the world recommends. It is not cautious. It is incoherent, while wearing caution’s clothes.

That is the sharpest thing in this part, and it is not an argument against caution. It is an argument that this particular formal apparatus does not survive contact with a result that says the pessimistic branch is not the same branch everywhere.

There is nothing to take a worst case of

Condition three has a second requirement that is simpler and even harder to get around. A worst case has to be finite. You need bounds.

The paper searched for them and reports what it found, which is nothing usable.

The loss per misuse event multiplies the misuse term identically under both readings, so the stakes per event are the same under each — which means the choice of reading cannot be the thing that makes one branch catastrophic. No bound on the health flow from the benefit was located. No bound on the probability that a candidate treatment succeeds was located.

Without bounds, there is no finite worst case to minimise against. “Assume the worst” is not a rule when the worst is unbounded; it is a refusal to choose, dressed as a choice, and it will always select whichever option forgoes the unbounded quantity — which here means always declining the benefit, for any benefit, forever.

The symmetry nobody has bounded

One more piece, and it is the one I would most like people to notice, because it is where intuition most reliably misleads.

The non-identification is symmetric between the two errors. The paper cannot sign the balance, which means it cannot tell you that acceleration is harmful and it cannot tell you that it is safe. Both errors are available.

What need not be symmetric is the *loss from acting on the wrong one*. Being wrong about safety might cost far more than being wrong about danger — that is the standard argument for precaution, and it is a serious one.

But nothing the paper read bounds that asymmetry either. So the paper assumes no asymmetry and supplies none. It does not claim the losses are symmetric. It declines to put a number on how asymmetric they are, because it has no basis for

one, and an unargued asymmetry is exactly the kind of thing that smuggles a conclusion into a framework.

If someone wants to argue for precaution here, this is the place to do it: supply the asymmetry, argue for it, and say where it comes from. That is a legitimate argument the paper cannot make for you and does not block.

A second thing the paper will not weigh

While we are listing refusals, there is a related one from Part 4 worth restating here.

The timing argument for haste on health grounds relies on discounting the future — Part 4 showed that all 24,483 healthy years of a two-year acceleration came from the discount rate. And a case for weighing long-run catastrophe heavily sits right beside it, resting on the opposite intuition about how to treat the future.

The paper does not weigh those two against each other. They are both legitimate positions about intertemporal choice, the choice between them is not an empirical question, and resolving it is not what a paper about mechanisms is for.

The levers, and why none of them is ranked

The paper discusses interventions. It ranks none of them, and the reason is structural rather than diplomatic: the model has no common resource variable, so there is nothing in which to express "this one is better per unit of effort", and the quantities a ranking would need are the unmeasured ones from Part 10.

Two of the levers can be read through the defence clock, and one of them produces the most counterintuitive result in the paper.

A countermeasure programme. Such a programme shortens the capability distance a countermeasure waits on, shortens its deployment time, and shifts how much of its wait is technical rather than institutional. Two things are then true at once, and they are easy to confuse.

At a fixed speed, shortening that distance lowers misuse loss — whenever the countermeasure is still waiting on capability and the hazard before it lands exceeds the residual after. That is simply what the programme is for, and it is the adoption floor from Part 4 seen from the other side.

But under the level reading with a fixed residual, a *longer* distance makes acceleration more likely to lower misuse loss. The further away the countermeasure, the more there is for speed to pull forward.

Put those together. **A countermeasure programme that succeeds lowers misuse loss at any given speed, and simultaneously weakens the case that going faster helps.** There is no contradiction; they are derivatives of different arguments, one holding speed fixed and one varying it. But anyone arguing "we should accelerate because countermeasures arrive sooner too" should notice that succeeding at countermeasures undermines their own argument for acceleration.

There is also a limit on the programme that the formal checks turned up and that I would not have predicted. A large enough cut to the countermeasure's distance takes it out of the regime where the result was derived: the clock becomes adoption-bound and further capability buys nothing. In the numerical check of a simpler unweighted version of the rule, **all 46 sign disagreements** were cases where the move had crossed onto the adoption branch. The same check shows that straight-line predictions fail when a move leaves its regime — which is a general warning about using a marginal result for a large intervention.

Oversight and interface screening. This lever moves the facility event rate or the hazard level. Its effect on the capability distances is not derived, so it does not enter the defence clock at all, and its magnitude is unmeasured.

What the structure does say is where it could act. Closing the gap in which funding sources trigger oversight changes *which facilities a regime reaches* — a question of coverage rather than of stringency, and the facility in Part 1 is a live example of why coverage is the binding question. And screening at the point

where a digital design becomes physical material is a distinct instrument from regulating a facility: different actors, different chokepoint.

One observation about the international layer, which is a statement about the text of the instruments rather than about their effect. It imposes purpose-based prohibitions and domestic prevention duties, and it contains **no facility-level verification system**. Nothing the paper read measures what those duties achieve.

So the levers come out as conditions, each with a named deciding quantity, and each of those quantities unmeasured. That is a thinner output than a ranking. It is also the shape of the honest answer, and it tells a reader which question to ask of any proposal: which quantity does this depend on, and does anybody know it?

Declining is not the same as shrugging

I want to separate two things that look alike from outside.

A shrug says the question is unanswerable, so do as you like. That would be a dereliction, and it would also be false — the paper establishes a good deal, and Part 4 is a list of things that are settled whichever way the hard choices go.

A declination says something narrower and more useful: *this particular output — a sign for the whole balance, a ranking of the levers, a decision rule — does not follow from what has been established, and here is exactly what would be needed to produce it.* That is a result with content. It tells you which arguments in circulation are unsupported, which is a service even when it is an unwelcome one.

The test I would apply: can a reader use it? From this paper a reader can check whether an argument they have just heard depends on a branch that the evidence does not pick, or on a quantity nobody has measured, or on a discount rate doing undisclosed work. That is usable. A manufactured recommendation would not be.

What the paper offers instead

Three things, in place of the recommendation.

The results that hold whichever way the choices go, from Part 4, each with its conditions — including the one that is most directly actionable, the adoption floor: if your countermeasure is stuck in approval rather than in the laboratory, accelerating the science is not a lever.

An explicit account of which conclusions depend on which choice, from Part 5, so that anyone who does hold a view about the branches can read their own answer off the paper rather than taking the author's.

And a list of what would settle what, which is Part 10 — the closest thing to a conclusion the paper has, and the part I would give to somebody with a research budget and an afternoon.

That is less than a recommendation. It is also, on this evidence, the most that can honestly be offered, and I would rather hand over a map with the unsurveyed regions marked than a route drawn across them.

Part 10 — What would settle what

This is the part I would hand to someone with a research budget, and it is the closest thing the paper has to a conclusion.

If the balance cannot be signed today, the useful question is what would change that. Not "more research" — the useless answer — but which specific quantities, measured how, would move which specific result. The paper ends with that list, because a non-identification result is only worth having if it tells you where to dig.

The quantities, and what each decides

The paper names roughly twenty quantities its results depend on and that the documented search supplies in none of its units. The important ones, with what each decides.

The health flow from the benefit — how many healthy years a year an arriving treatment actually delivers. This sits in front of the entire benefit side. Without it there is no benefit term, only a benefit-shaped placeholder.

The loss per misuse event. Multiplies the misuse term under both danger readings identically. Nothing in Part 7 bounds it in either direction, and it is the quantity that would make any precautionary argument tractable.

The risk per unit of capability — the quantity the whole field is implicitly arguing about, and for which the expert consensus the paper read says there is not yet a standard.

The capability-set share of a countermeasure's wait. How much of the gap between "someone could build this" and "patients have it" is technical rather than institutional. This decides the adoption floor of Part 4, and it is the one item on this list that could plausibly be estimated from existing records.

The hazard elasticity — how sharply danger responds to the rate of advance. The deciding quantity under the speed reading, and absent entirely under the level reading.

The health cost per hospital attack, in population units. Part 8's factor of 260. No conversion of any hospital result into healthy years was located anywhere.

And the specification question itself: whether misuse risk is set by the capability level reached or by the speed of approach. Not a number but a modelling fact, and the one that decides most of the rows in Part 5.

The paper's own figure maps eight of these against the part of the argument each would decide. Worth being clear about what the map is: it states the paper's dependencies and supplies none of the quantities.

Four observations that would move them

The list above is quantities. This is the shorter and more useful list: four things somebody could actually go and do.

One. A sensitive bound on misuse-relevant assistance, from a design much larger than twelve cells. Part 7's controlled comparison could detect about one point on a nine-point scale and observed 0.22. A design large enough to bound the effect near zero — or to find it — is the single highest-value study in this literature, and the obstacle is not conceptual. It is sample size, and the cost and the access that sample size would require.

Two. A within-programme estimate of development time under differing availability of general capability. Take real drug development programmes and estimate how much faster the same programme runs with more capable tools available. This is the benefit side's missing number, and it is the kind of thing firms may already have internally without anyone having published it.

Three. A translation of the hospital result into healthy years. This needs one thing: the remaining healthy life expectancy of the people who actually died during those attacks. Not the mean age of inpatients, which is what exists. The decedents' own ages and prognoses. That single addition would collapse Part 8's factor of 260 into something usable, and the data to do it plausibly exists in the claims records the original study already used.

Four. For the defence clock: a dated record placing a countermeasure's capability requirement against the corresponding misuse requirement. Part 6's entire result is a comparison of two capability distances, and nobody has ever written down a matched pair. Even a handful of historical cases — this countermeasure needed roughly this much capability, the corresponding danger needed roughly that much — would turn the defence clock from an unusable inequality into a testable one.

Of the four, the third is the cheapest and the first is the most valuable. None requires a conceptual advance.

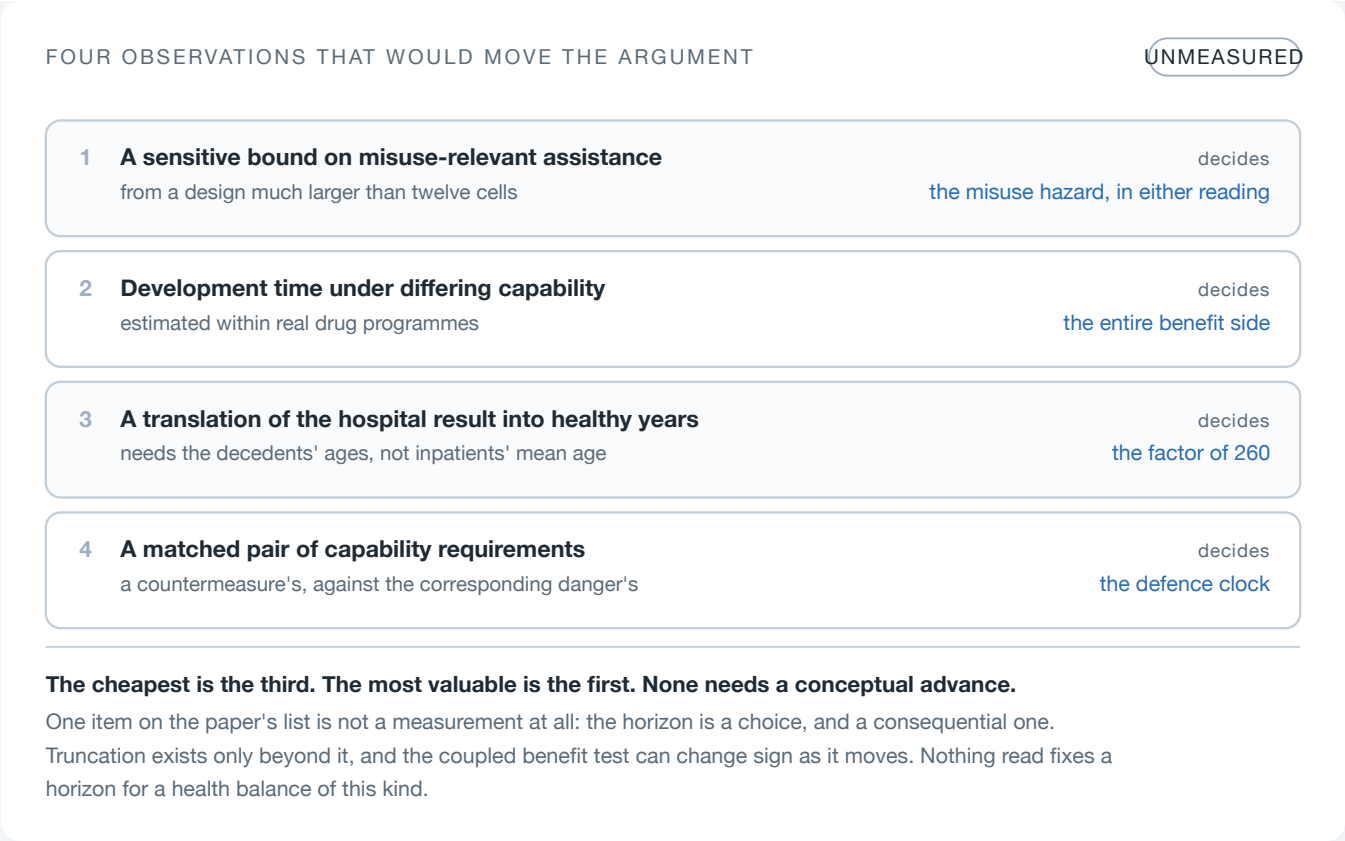


Figure 22. The closest thing the paper has to a conclusion, and the part to start from if you want to improve on it. From the paper's estimand list.

One item on that list is not a measurement at all

The horizon — how far into the future the ledger counts — is not a gap. It is a choice, and the paper makes it, and it is consequential enough that it belongs in this part rather than buried in the set-up.

Truncation from Part 5 exists only where the dangerous capability lies beyond the horizon's reach at the current speed. Move the horizon and the mechanism switches on or off. And on either version of the speed reading, the test for whether a coupled benefit is worth it can change sign as the horizon moves.

Nothing in the documented search fixes a horizon for a health balance of this kind. There is no convention to appeal to. So the paper states its results for a horizon taken as given, and flags the horizon on its own map as what it is: a decision, not a missing measurement.

This is worth generalizing. In any analysis of this shape, the window is doing work, and an analysis that does not say what its window is has not told you one of its main assumptions.

What the search actually was

Numbers, so that “the documented search” means something.

Seven literatures, searched on 26 October 2026. 173 source records covering 162 distinct works — a work read in two strands counts twice as a record and once as a work.

The read-depth split is the part I would want a critic to see. Of the records, **22 were read in full, 113 in part, 17 at abstract only, and 21 were not obtained at all.** A work the paper could not obtain appears in it only as an item it tried and failed to obtain, never as evidence. Findings resting on abstracts or on secondary accounts are labelled as such wherever they appear.

Twenty-two read in full out of 162 is a real limitation and the paper prints it rather than writing “a comprehensive review of the literature”. Throughout, “unmeasured” means that no measure was located in that search — a statement about the search, not about the world.

What the paper refused to read, and what that costs

On safety grounds the paper did not collect, read for, or record technical detail about how biological harm could be caused, and excluded exploit detail on the cyber side.

The consequence is specific and I would rather state it than let a reader discover it. The paper can say what a study measured, how well it measured it, who funded it and what the field concludes. It cannot say which agents or methods any capability bears on. No part of its risk term rests on such material, which is a design choice rather than an accident — the terms were built so that the excluded material would not be load-bearing.

One source was kept as a safe excerpt only: its design, its authors' stated limits, its governance proposals. The one controlled comparison was read the same way — design, sample, scoring rubric and statistics taken, and its section reporting model outputs never opened.

What was excluded was operational detail, not risk estimates. Nothing was dropped because it was inconvenient.

What a safety classifier stopped

An unusual disclosure, included because leaving it out would misrepresent how the paper was made.

The work was done with large language models, and four steps were stopped by their safety systems.

One was a summary of the reviewers' methodological criticism of how incident, containment and evaluation statistics had been used in this literature. It was stopped four times across two different models, including after rephrasing. The paper did not re-derive it: the review reports stayed on disk and were applied by reference to their numbered findings.

A second step was stopped once and resumed at the structural level only. A third — drafting the misuse-evidence section — was stopped when the statistics-only extraction from the controlled comparison was held in the same working context as the summary of the misuse evidence; that section was then drafted from the summary alone, separately from the other evidence sections. A fourth was stopped while planning the refusal section of Part 9, in a step that read across the misuse and the facility material at once; no draft text existed yet, and both sections were then written at the level of governance, oversight, study design and model structure only.

The paper's judgement, which I stand behind: the division of material was deliberate, and it cost the paper no content. But a reader is entitled to know that

the research process had constraints of this kind and to form their own view about whether the fourth stop, in particular, shaped what got written.

Ordinary gaps

Not every limitation is interesting. These are the dull ones, listed because a limitations section that contains only interesting limitations is not a limitations section.

The published text of the one causal hospital-mortality study was closed to the paper, so its abstract and its working paper are reported separately and never spliced together. Incident updates from one national health service were blocked to automated fetching. A laboratory-safety manual and the current US synthesis-screening rules could not be confirmed from primary documents. Two studies on the benefit side were paywalled. The 2026 edition of the international safety report was not read — so the absence the paper records in that literature speaks only to the editions actually in its search logs.

What the paper actually contributes

To close, the honest summary of what a hundred pages bought.

It put four channels — benefit, misuse, facility events, health-system attacks — into one unit, with a rule ensuring no year of health is counted twice, and showed that the rule is an accounting identity rather than a finding.

It identified three modelling choices that the literature does not settle and that are not gaps a longer search would close, and carried both sides of each rather than picking.

It sorted its results into those that survive all three choices — five mechanism results plus the identity, each within its conditions — and those that do not, labelling each conditional result with the choice it depends on.

It found that its own most striking result, the four opposite signs, drops to one when the comparison is made consistently; and that its most invariant-looking result is invariant in form and not in magnitude. Both of those corrections made the paper weaker and were made anyway.

It showed that the one channel with a measured causal magnitude is the one channel not empirically linked to AI capability, and that a reassuring reading of the evidence therefore reflects measurability rather than safety.

It declined to produce a recommendation, and said why the obvious precautionary rule cannot be applied to this structure.

And it ended with a list of what would settle what, which is where anyone wanting to improve on it should start.

The division of the deployment wait into a capability-set part and an adoption-set part, and the floor that follows from it, is the piece I would most like to be used by somebody else. Everything around it is scaffolding for a question that remains open.

Want to learn more?

The technical paper. Everything in this guide comes from *Running Toward the Clinic: Speed, Biological Risk and Health in the AI Transition*, and from its appendix and supplement. The paper carries the full reference list, the proofs, the twenty-seven figures and the record of how each result was derived and checked. Where this guide gives a number, the paper gives its source.

Its predecessor. *Running Through the Rain: Speed, Secrecy and Risk in the AI Cyber Transition* asks the same question about cyberspace rather than biology, and has its own plain-language edition. Part 1 explains what changed between them.

The sources this guide leans on most. I am naming only the ones I can name precisely, and pointing at the technical paper for the rest, because a reading list

with a wrong entry in it is worse than a short one.

- *The one controlled comparison* — Mouton and colleagues, 2024: twelve cells of three researchers, randomised across four scenario strata, plans scored on a nine-point viability scale. Part 7 describes the design, the null, and the authors' own sensitivity figure. Their own statement of their limits is the most useful sentence in the literature.
- *The hospital-mortality study* — a stacked event study and admission-level difference-in-differences on Medicare claims, 74 attacks on 163 hospitals, 2016–2021. It exists in a published version and a working paper that do not agree on magnitude; Part 8 reports both and says which was read in full.
- *On when "assume the worst" is a defensible rule* — John Rawls, *A Theory of Justice* (1971), §26, pages 153–154, where the three conditions are set out and the rule is described as a maxim for special circumstances rather than a general guide. Stephen Gardiner's 2006 paper adopts the same three conditions for a core precautionary principle and states that they are sufficient rather than necessary. Part 9 checks this problem against all three.
- *The virology benchmark* and *the expert-consensus review of the field's evaluation methods* are described in Part 7 and cited in full in the technical paper.

Where the numbers in this guide come from

Four kinds of number appear in these ten parts, and they are not equally solid. The guide tries to say which is which every time, and this is the key.

Published findings. The 94 per cent benchmark result, the arm means and the t-statistic from the controlled comparison, the hospital study's percentage-point effects on their own baselines, the California volume figures, the rise in attacks from 43 to 91 a year. These are other people's results, reported as they printed them.

The paper's own arithmetic on published numbers. About 0.6 to 0.9 excess deaths per attack. The reconstructed interval of about −1.5 to +1.0 viability points. The factor of 260. All three are clearly labelled in the paper, two of them graded WEAK by the paper itself, and none of them is the original authors'.

Worked illustrations. The plateau example, the 0.833 boundary, the 0.48 elasticity threshold, the discounted 24,483 healthy years. These use the paper's illustrative parameter values or standard conventions to show a shape. They are arithmetic, not measurement, and no empirical claim rests on them. I have checked each one rather than reasoning about it loosely, because a worked example with a mistake in it is worse than no example.

Counts about the work itself. 173 source records, 162 distinct works, 22 read in full. Seven literatures, searched on 26 October 2026.

If you remember one thing from this guide, let it be the distinction between the second and third kinds. A factor of 260 computed from real data is a statement about how little we know. A worked example is a statement about how a mechanism behaves. Neither is a measurement of the danger, and the paper contains no measurement of the danger, because nobody has made one.

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